

Russian trolls on Twitter

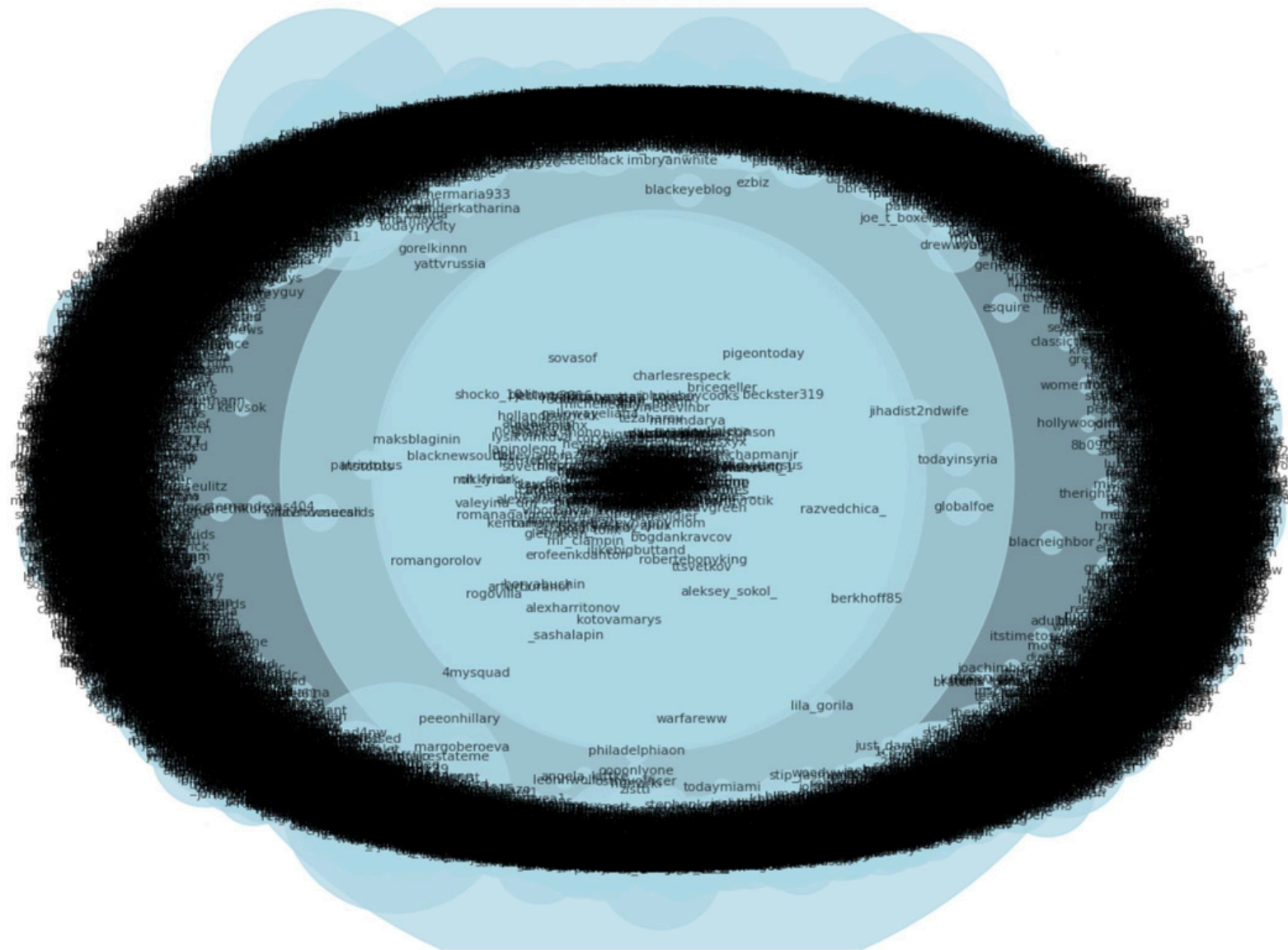
Bartosz Maj

Initial dataset

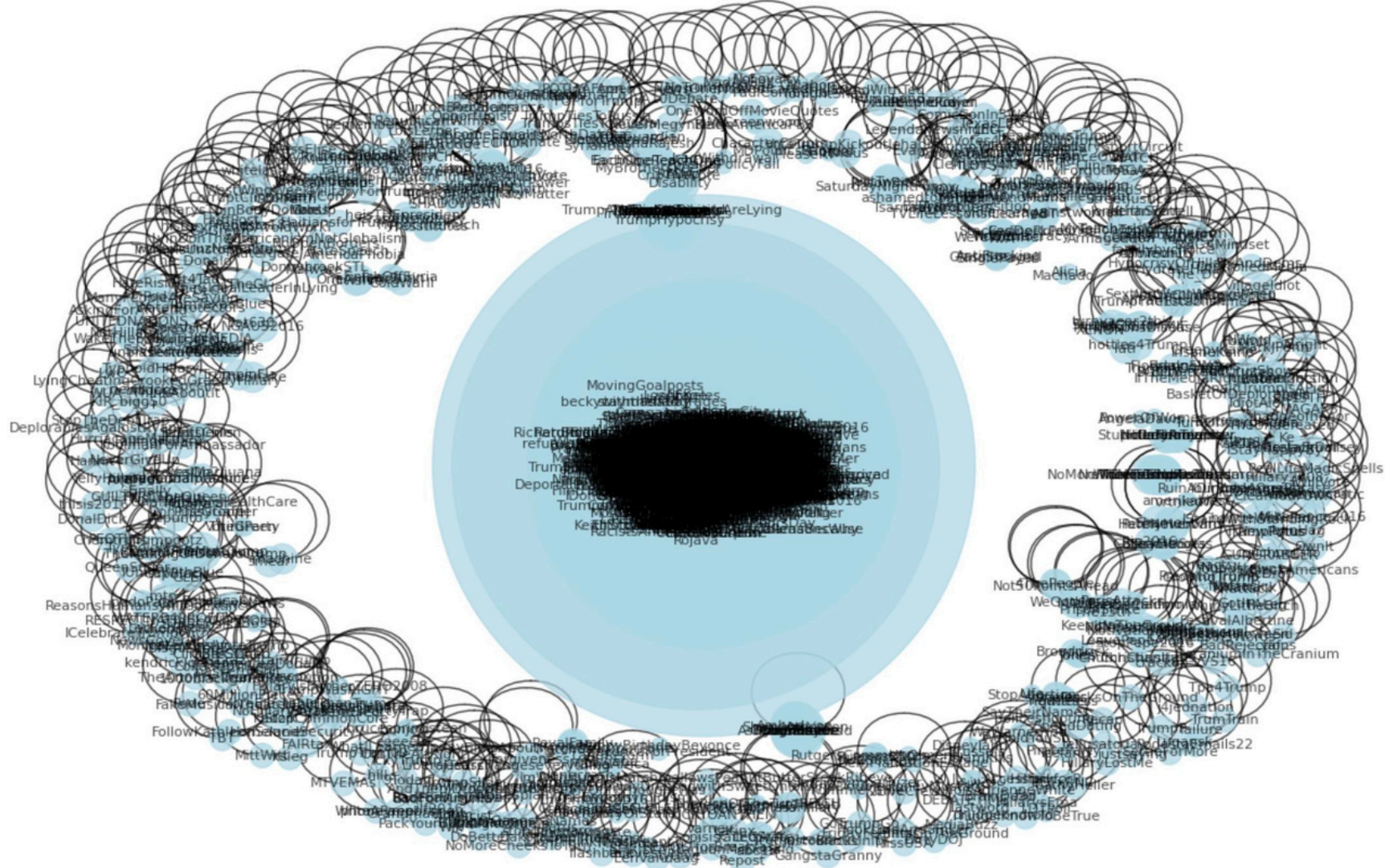
```
http://nodeassets.nbcnews.com/  
russian-twitter-trolls/tweets.csv
```

```
http://nodeassets.nbcnews.com/ ☒  
russian-twitter-trolls/users.csv
```

Initial graphs

[illegible]

Hashtag Co-occurrence Network



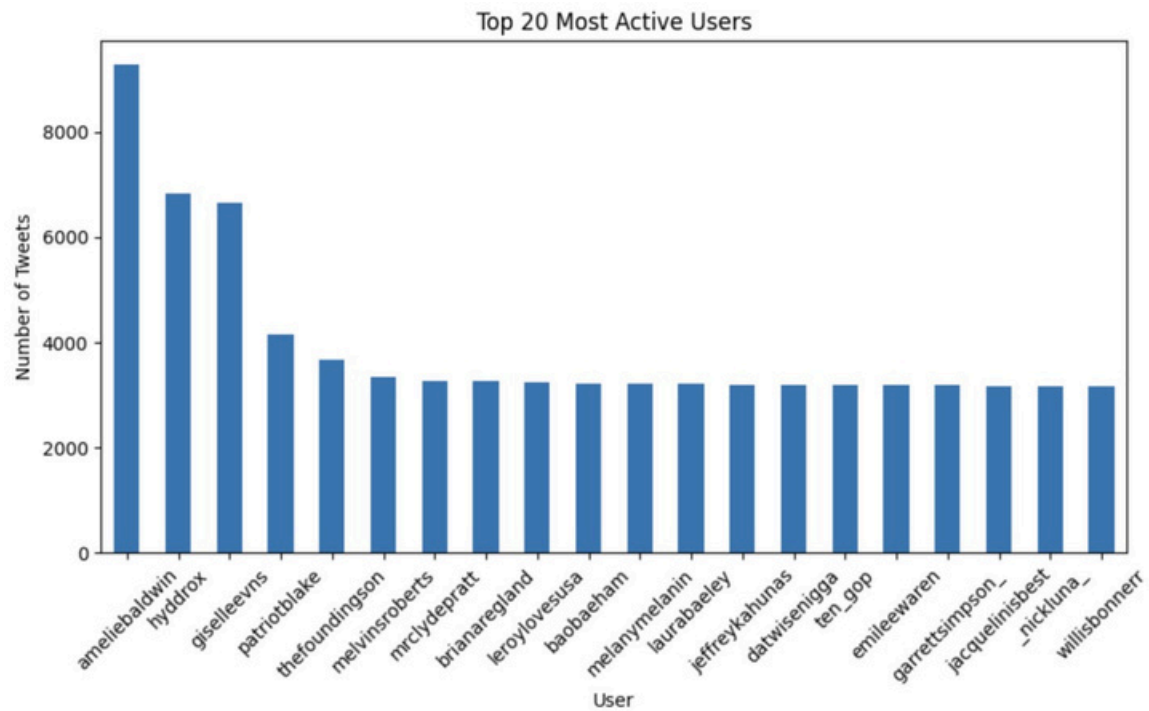
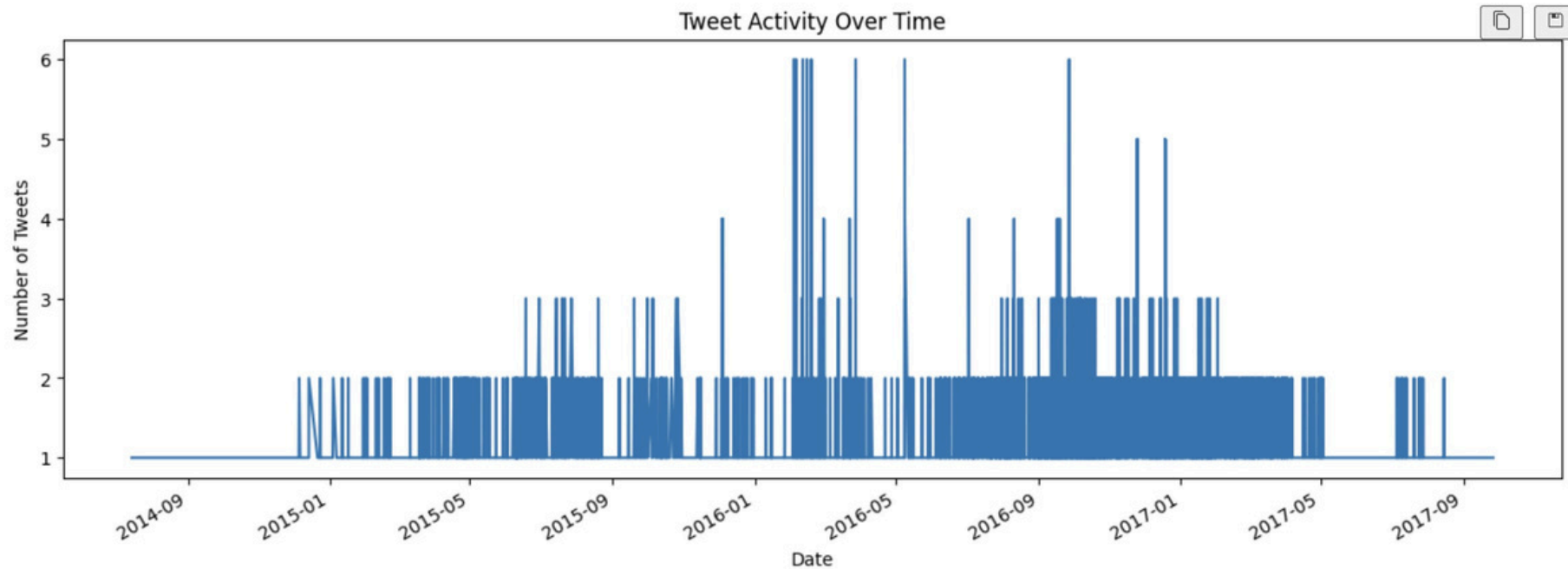
Mention Network Metrics:

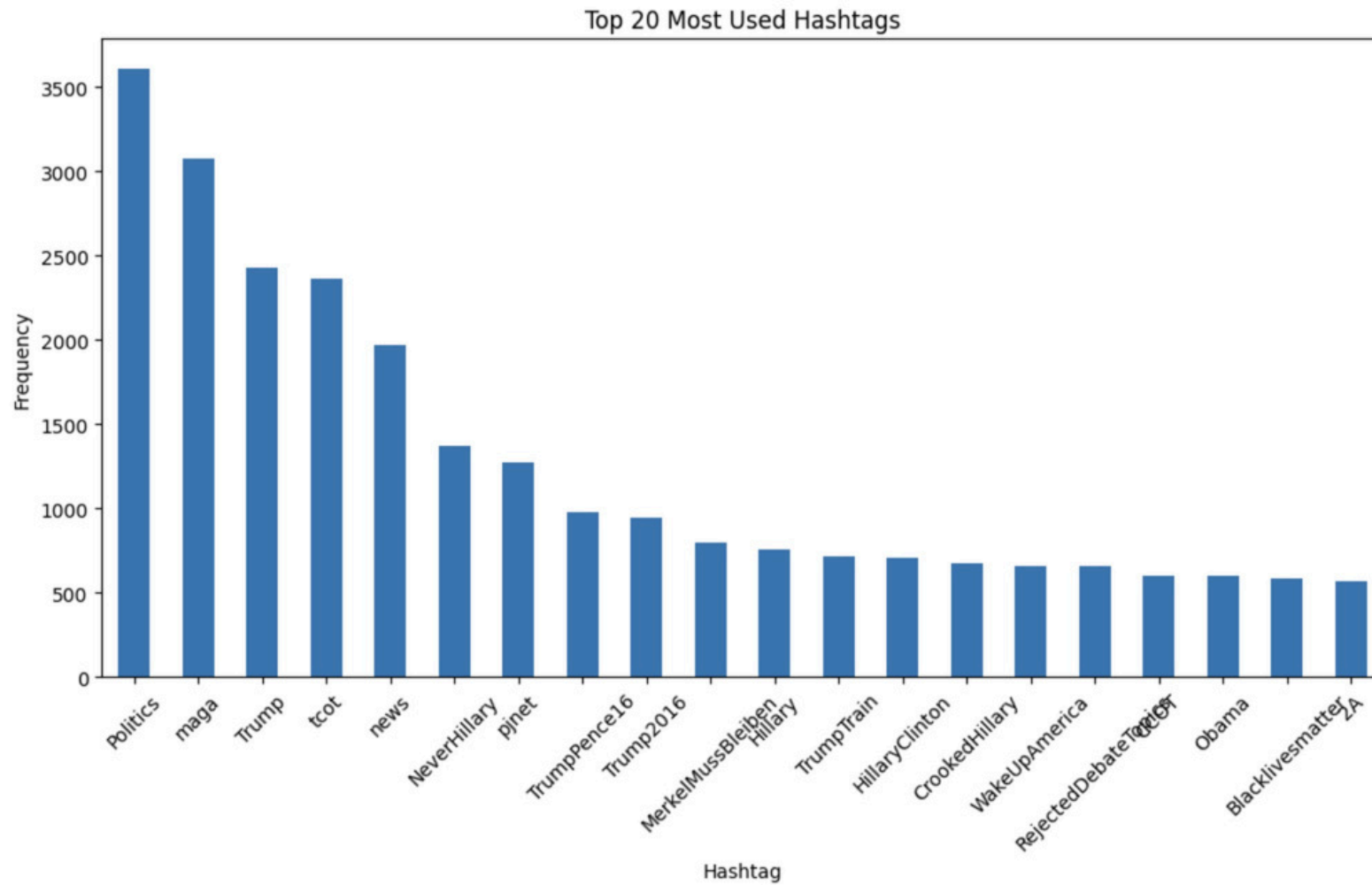
```
num_nodes: 14153
num_edges: 33180
density: 0.00016565710030696192
avg_in_degree: 2.344379283544125
avg_out_degree: 2.344379283544125
largest_scc_size: 12
largest_wcc_size: 14128
max_degree_centrality: 0.28158564160542676
diameter_calculation_error: DiGraph not strongly connected.
```

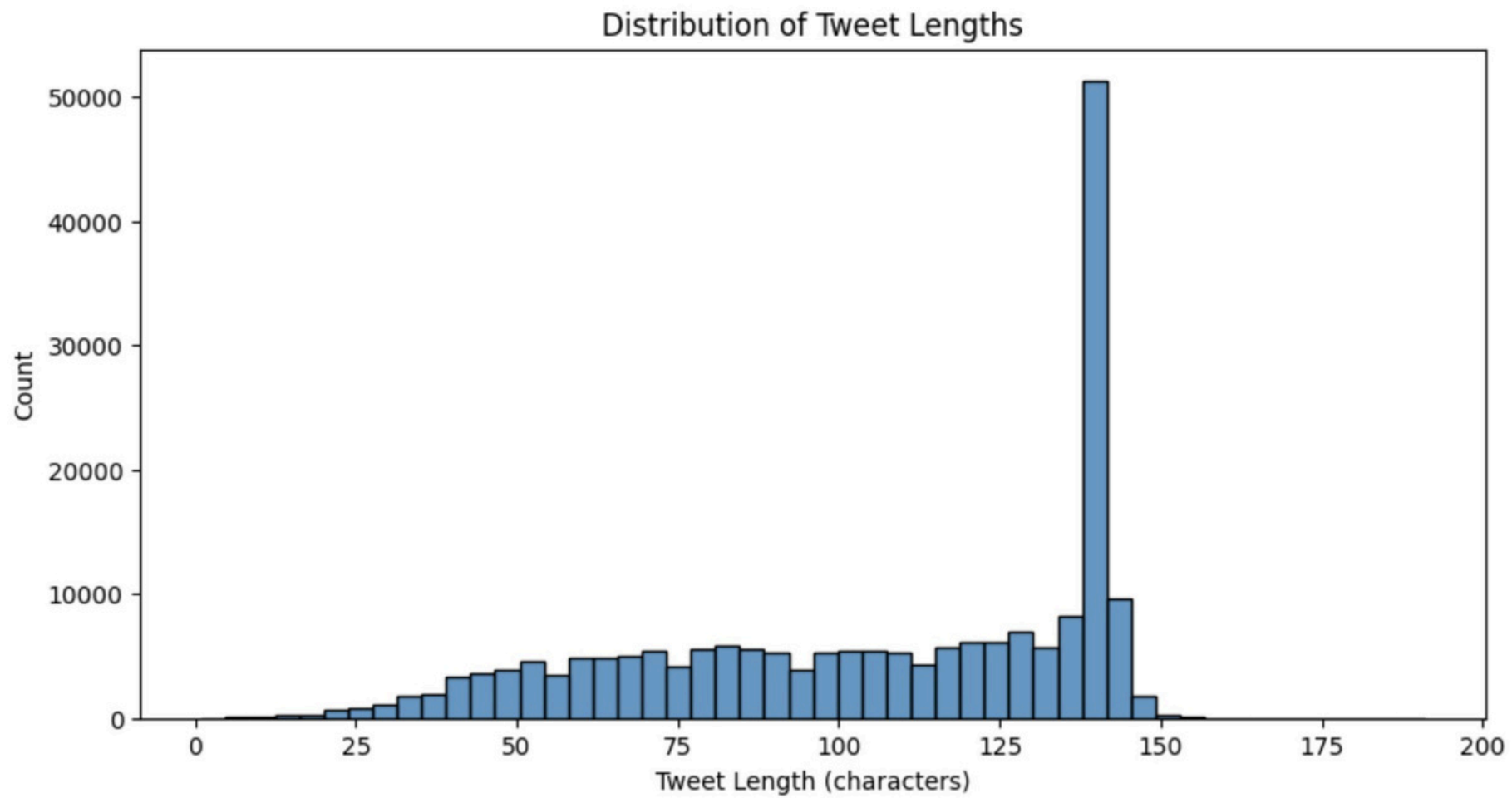
Top users by PageRank:

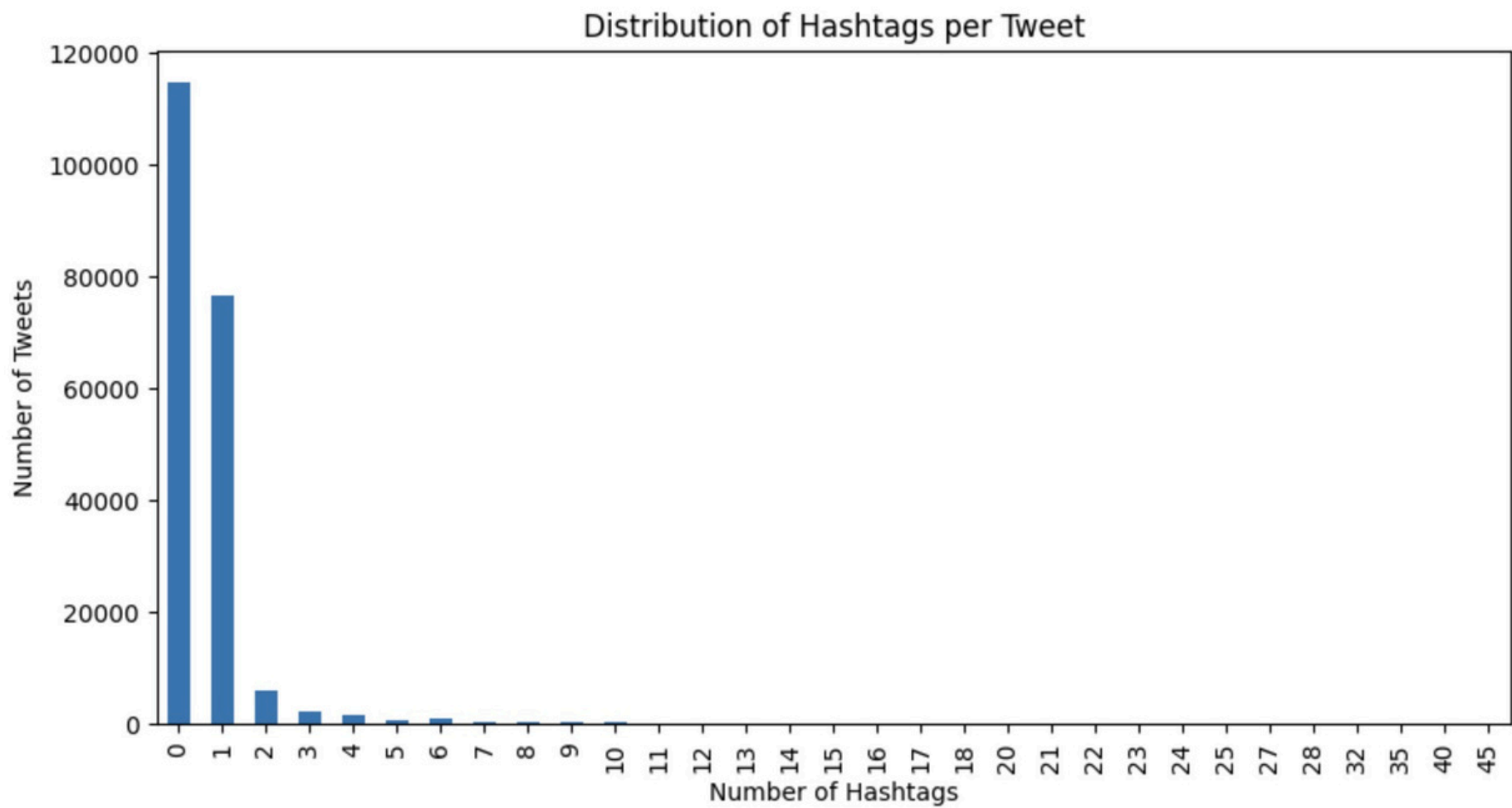
```
realdonaldtrump: 0.0012
romaacorn: 0.0007
hillaryclinton: 0.0007
tina_kandelaki: 0.0004
rt_america: 0.0004
potus: 0.0003
rt_com: 0.0003
wikileaks: 0.0003
tagesschau: 0.0002
yasyueksel: 0.0002
```

Initial EDA



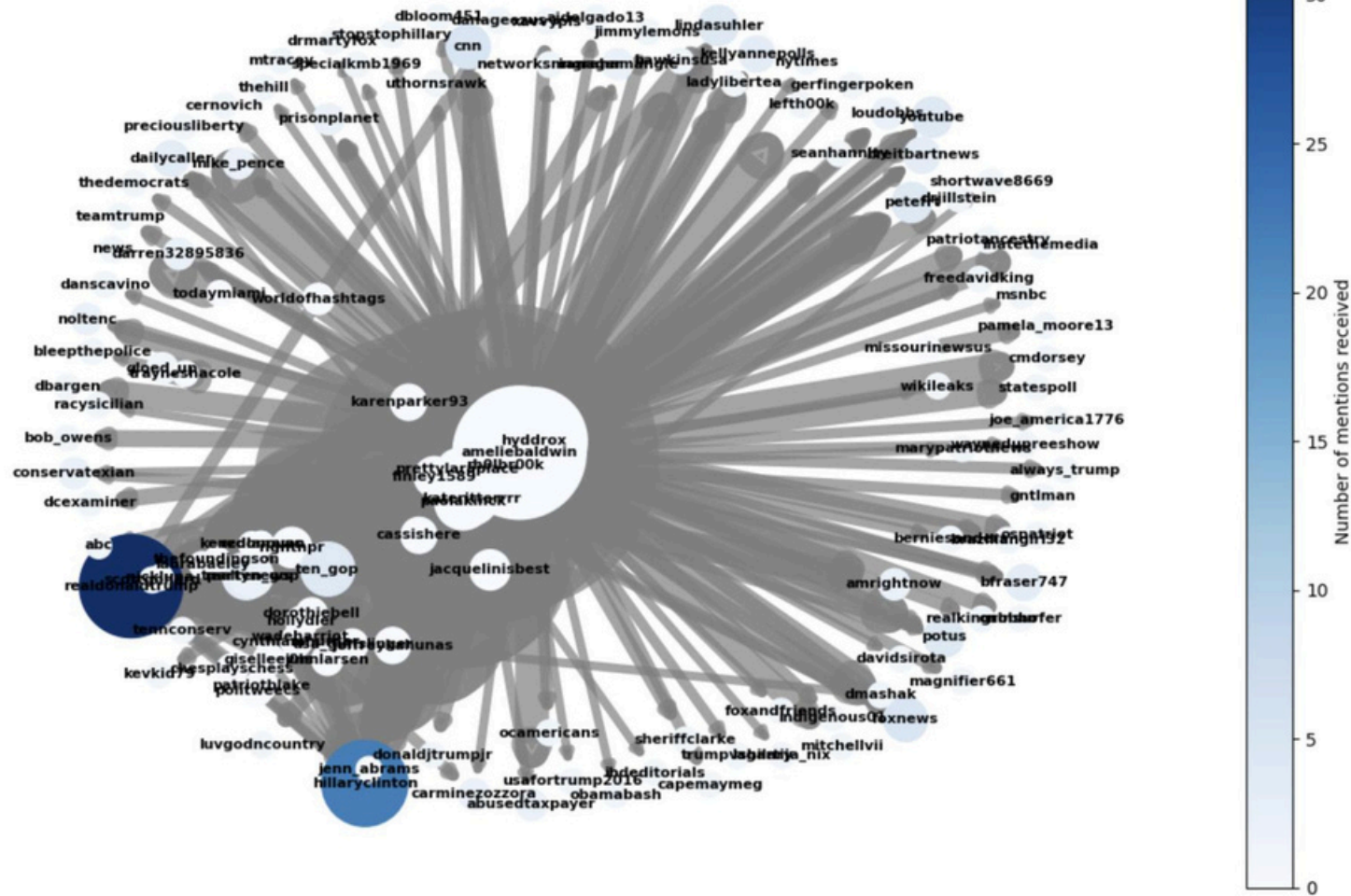






Mention networks in time

Mention Network - Period: 2016Q3



Network Statistics for 2016Q3

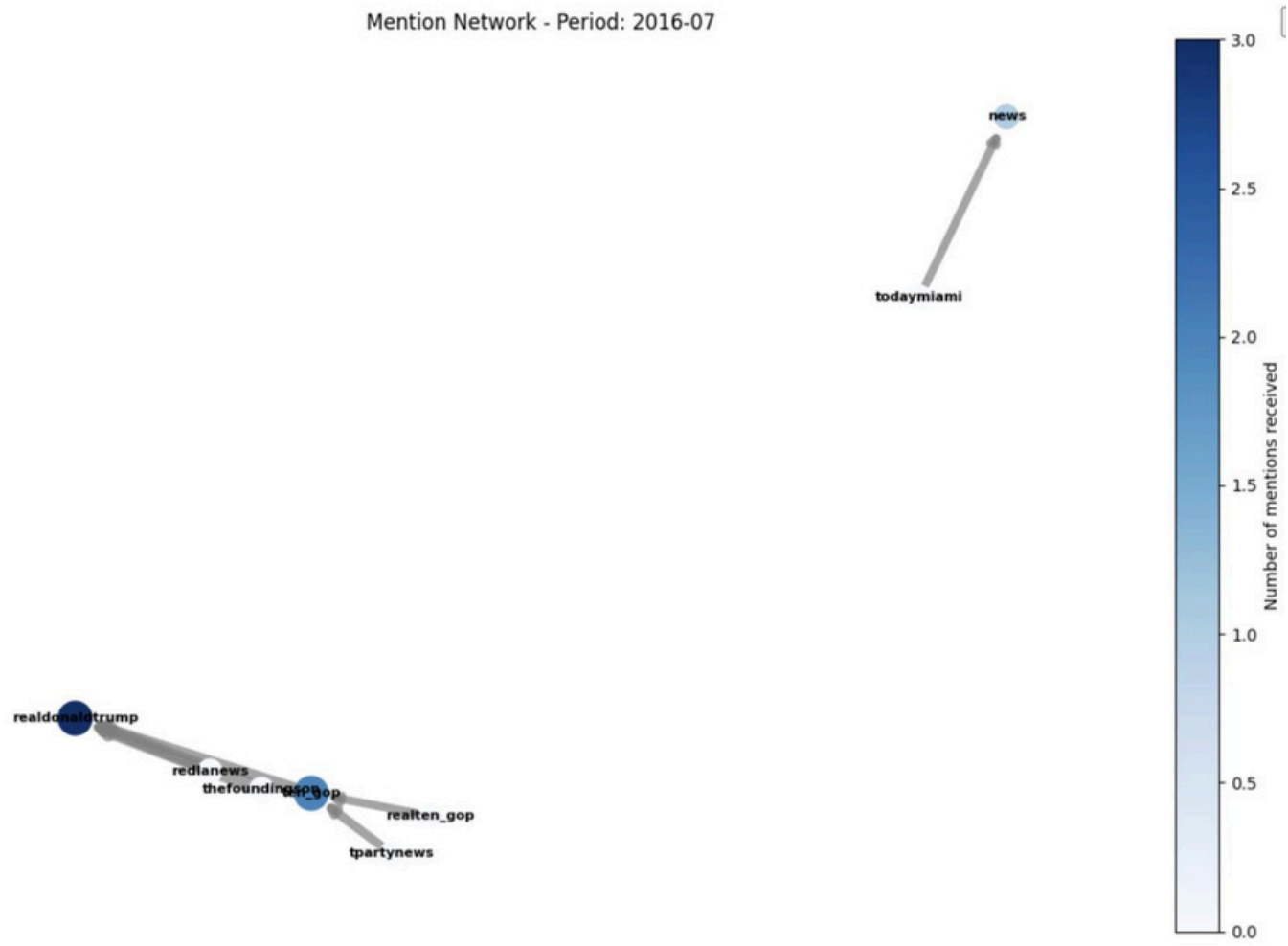
Number of users: 127

Number of connections: 190

Top 5 Users Mentioning Others:
 ameliebaldwin: 54 mentions made
 hyddrox: 35 mentions made
 rh0lbr00k: 16 mentions made
 paolakink: 9 mentions made
 kateritterrr: 8 mentions made

Top 5 Most Mentioned Users:
 realdonaldtrump: 32 mentions received
 hillaryclinton: 22 mentions received
 cnn: 5 mentions received
 foxnews: 5 mentions received
 youtube: 4 mentions received

Mention Network - Period: 2016-07

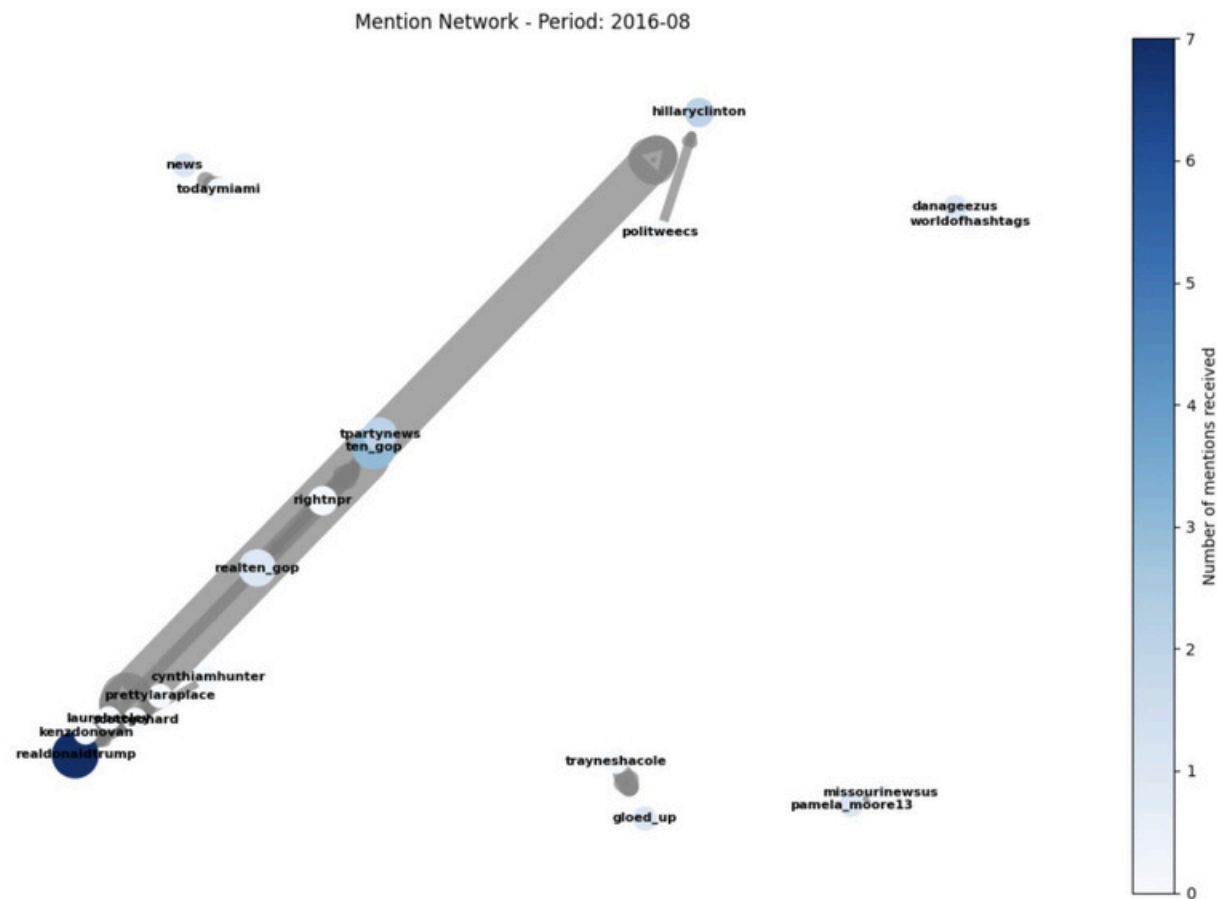


Network Statistics for 2016-07

Number of users: 8
Number of connections: 6

Top 5 Users Mentioning Others:
redlanews: 1 mentions made
tpartynews: 1 mentions made
ten_gop: 1 mentions made
thefoundingson: 1 mentions made
realten_gop: 1 mentions made

Top 5 Most Mentioned Users:
realdonaldtrump: 3 mentions received
ten_gop: 2 mentions received
news: 1 mentions received
redlanews: 0 mentions received
tpartynews: 0 mentions received



Network Statistics for 2016-08

Number of users: 20

Number of connections: 19

Top 5 Users Mentioning Others:

`ten_gop`: 3 mentions made

`realten_gop`: 3 mentions made

`rightnpr`: 2 mentions made

`missourinewsus`: 1 mentions made

`politweecs`: 1 mentions made

Top 5 Most Mentioned Users:

`realdonaldtrump`: 7 mentions received

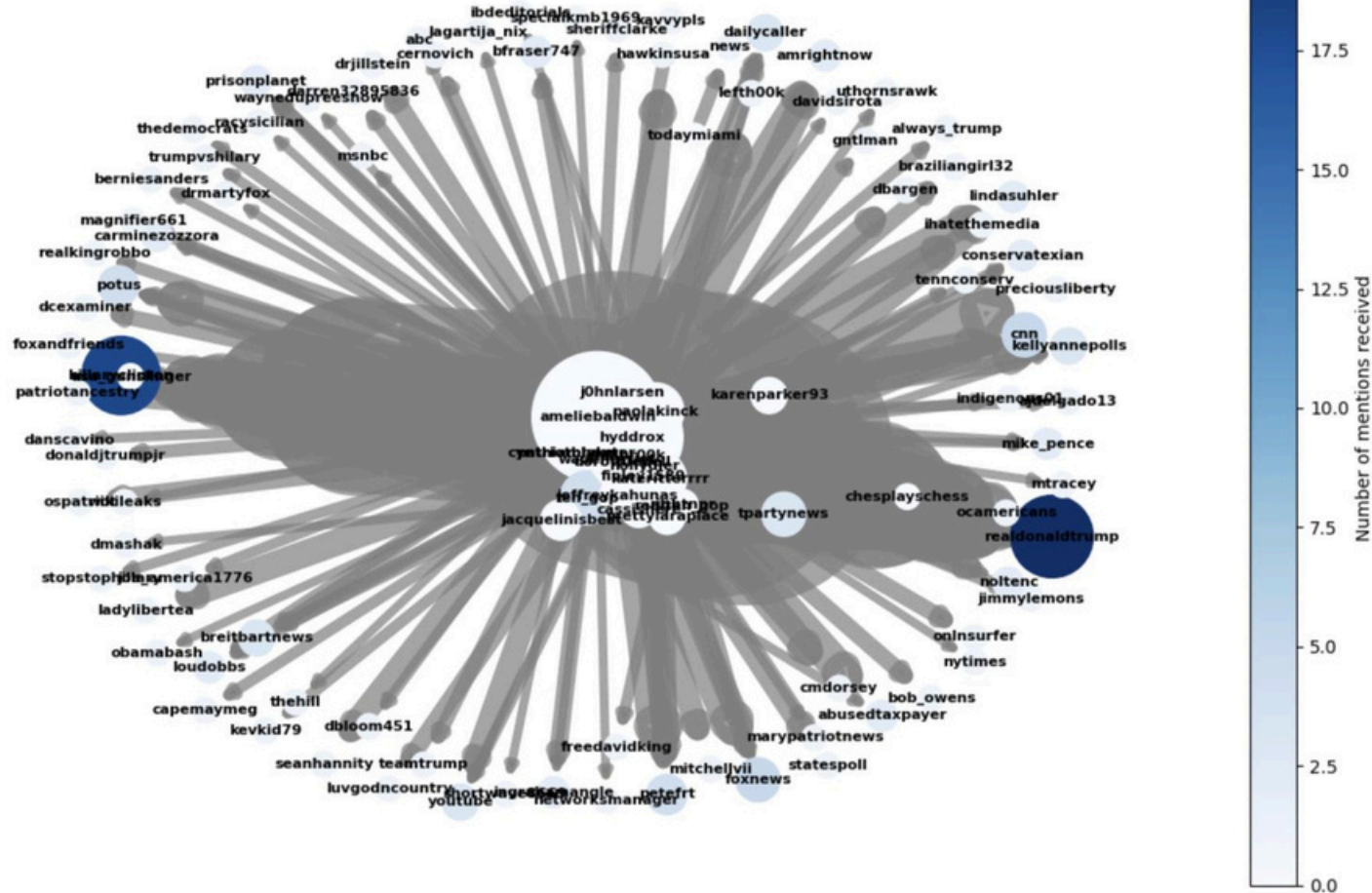
`ten_gop`: 3 mentions received

`hillaryclinton`: 2 mentions received

`tpartynews`: 2 mentions received

`pamela_moore13`: 1 mentions received

Mention Network - Period: 2016-09



Network Statistics for 2016-09

Number of users: 109

Number of connections: 161

Top 5 Users Mentioning Others:

ameliebaldwin: 52 mentions made

hyddrox: 32 mentions made

rh0lbr00k: 15 mentions made

paolakinck: 9 mentions made

kateritterrrr: 8 mentions made

Top 5 Most Mentioned Users:

realdonaldtrump: 20 mentions received

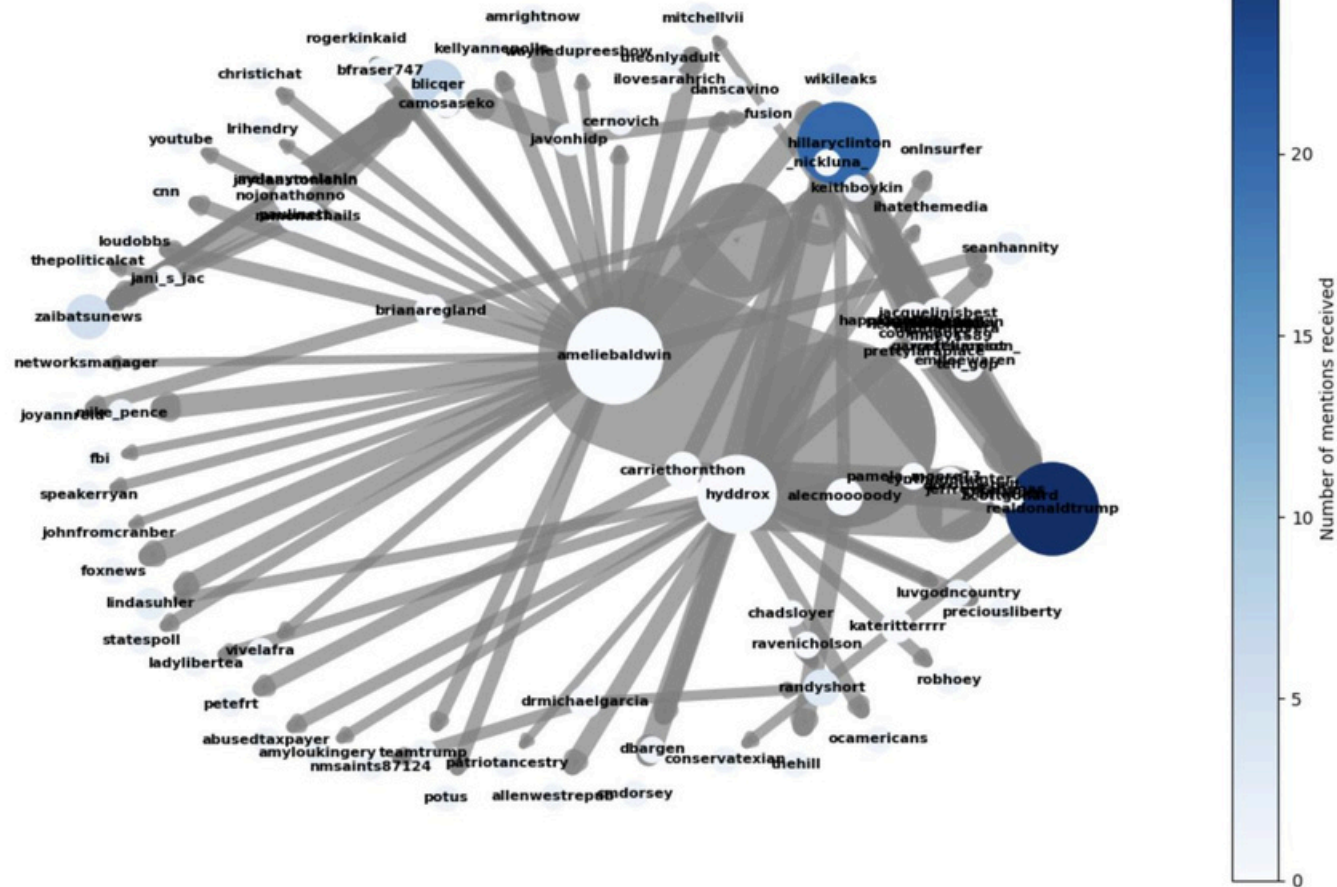
hillaryclinton: 18 mentions received

cnn: 5 mentions received

foxnews: 5 mentions received

ten_gop: 4 mentions received

Mention Network - Period: 2016-10



Network Statistics for 2016-10

Number of users: 93

Number of connections: 113

Top 5 Users Mentioning Others:

ameliebaldwin: 28 mentions made

hyddrox: 18 mentions made

carriethornton: 3 mentions made

patriotblake: 3 mentions made

alecmooooody: 3 mentions made

Top 5 Most Mentioned Users:

realdonaldtrump: 26 mentions received

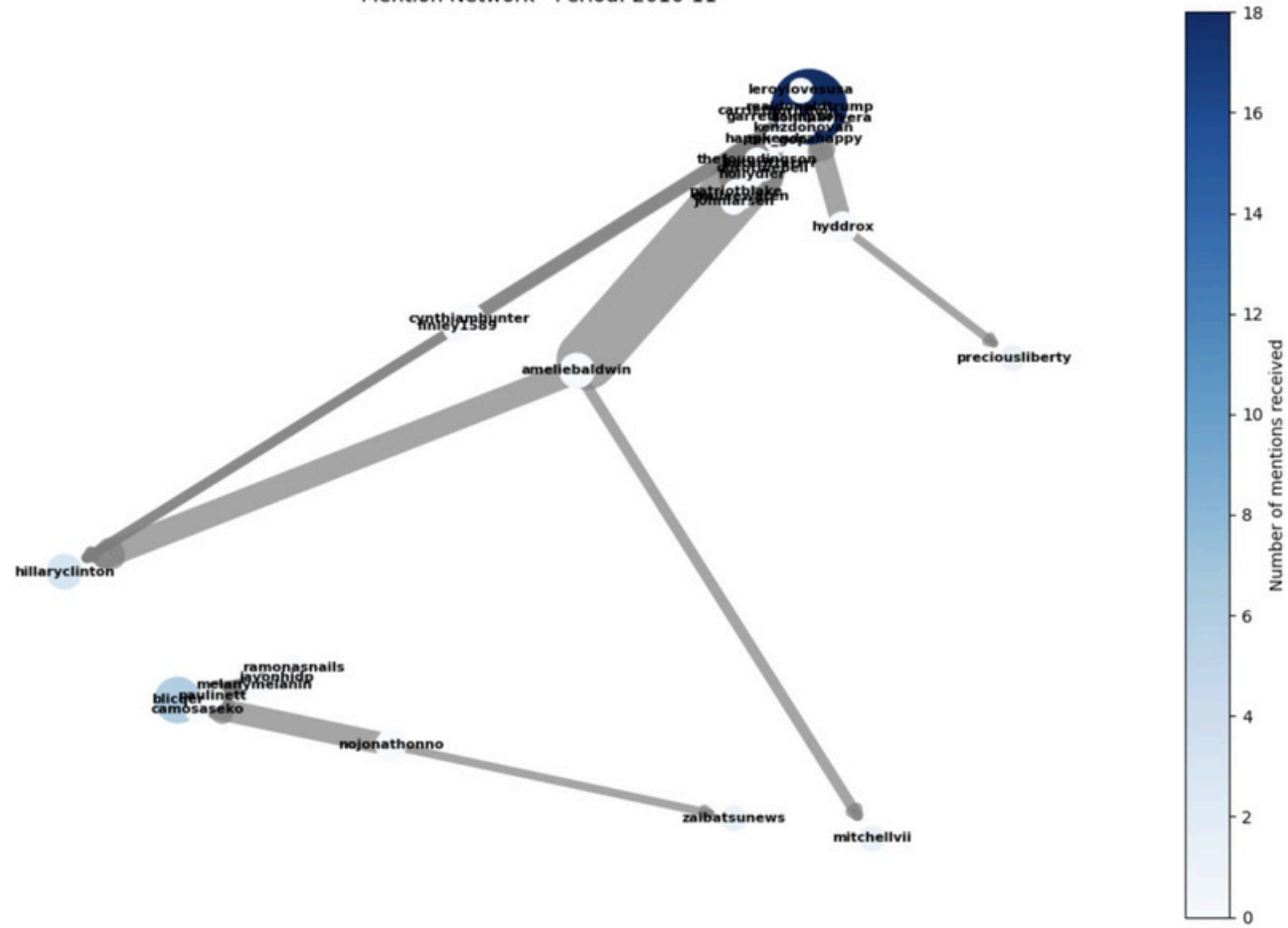
hillaryclinton: 20 mentions received

blicqer: 7 mentions received

zaibatsunews: 5 mentions received

randyshort: 3 mentions received

Mention Network - Period: 2016-11



Network Statistics for 2016-11

Number of users: 30

Number of connections: 30

Top 5 Users Mentioning Others:

ameliebaldwin: 3 mentions made

cynthiamhunter: 2 mentions made

hyddrox: 2 mentions made

nojonathonno: 2 mentions made

finley1589: 2 mentions made

Top 5 Most Mentioned Users:

realdonaldtrump: 18 mentions received

blicqer: 6 mentions received

hillaryclinton: 3 mentions received

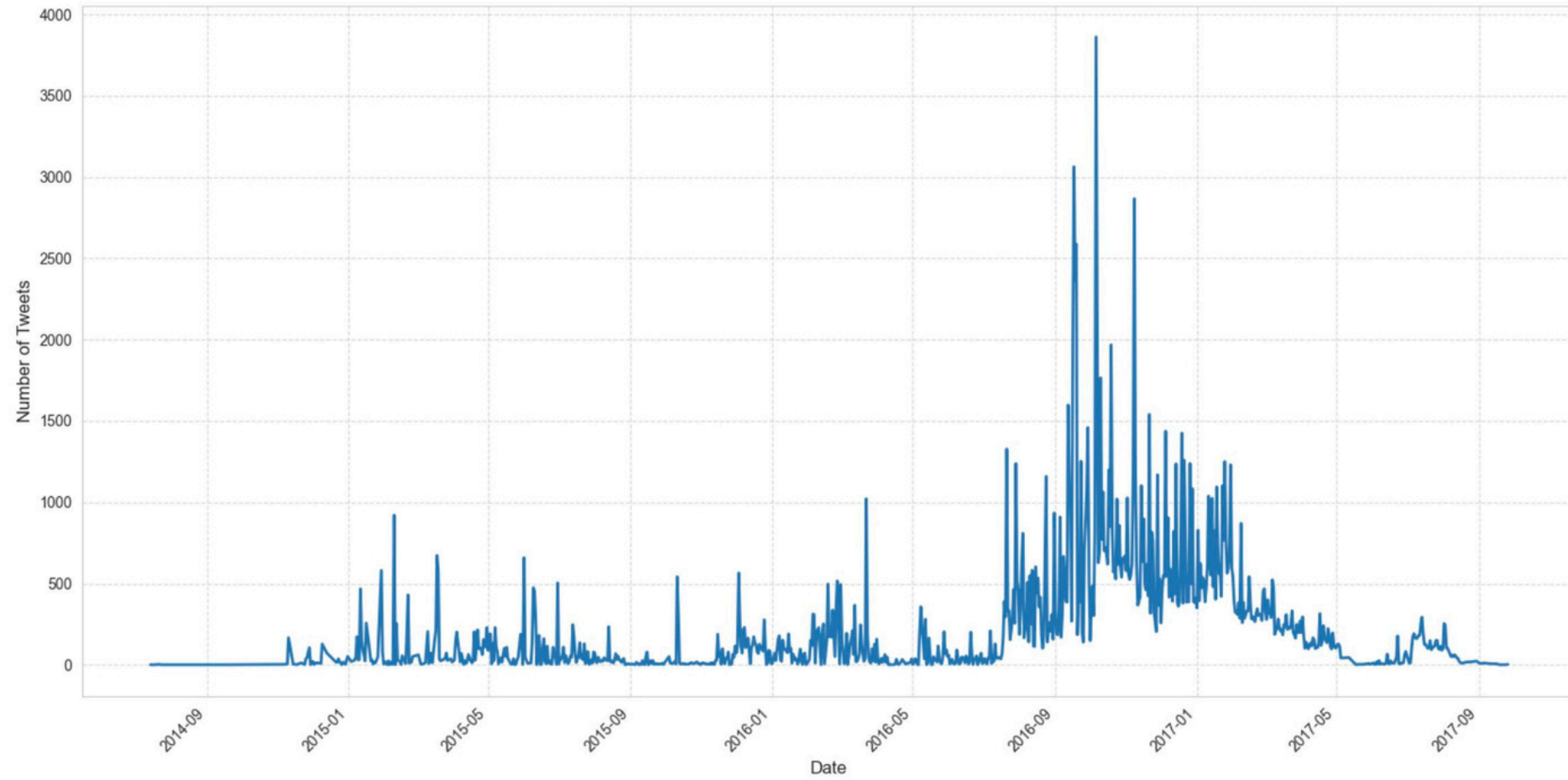
zaibatsunews: 1 mentions received

preciousliberty: 1 mentions received

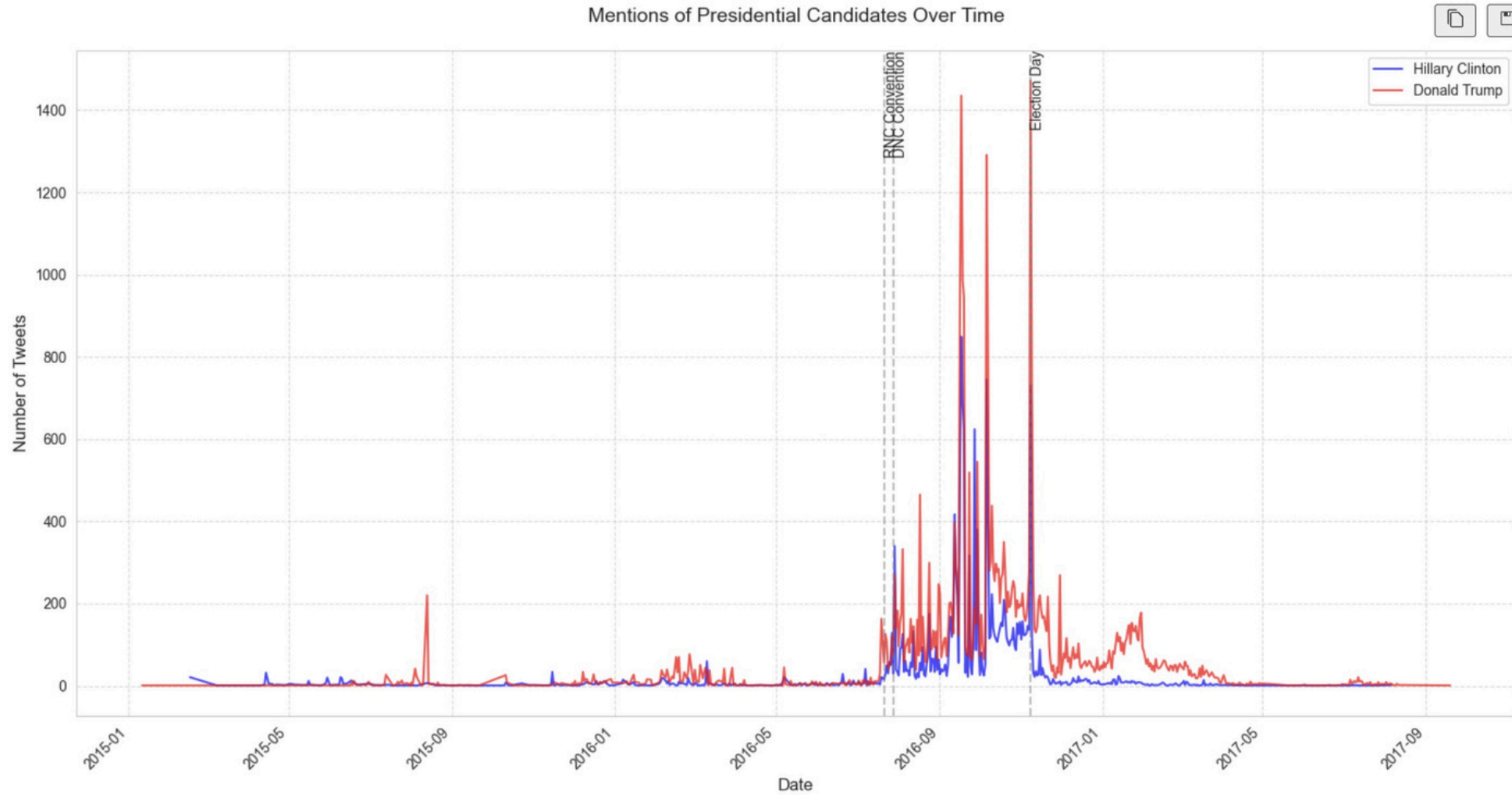
Strong correlation with politics?

(2016 US election)

Number of Tweets Over Time

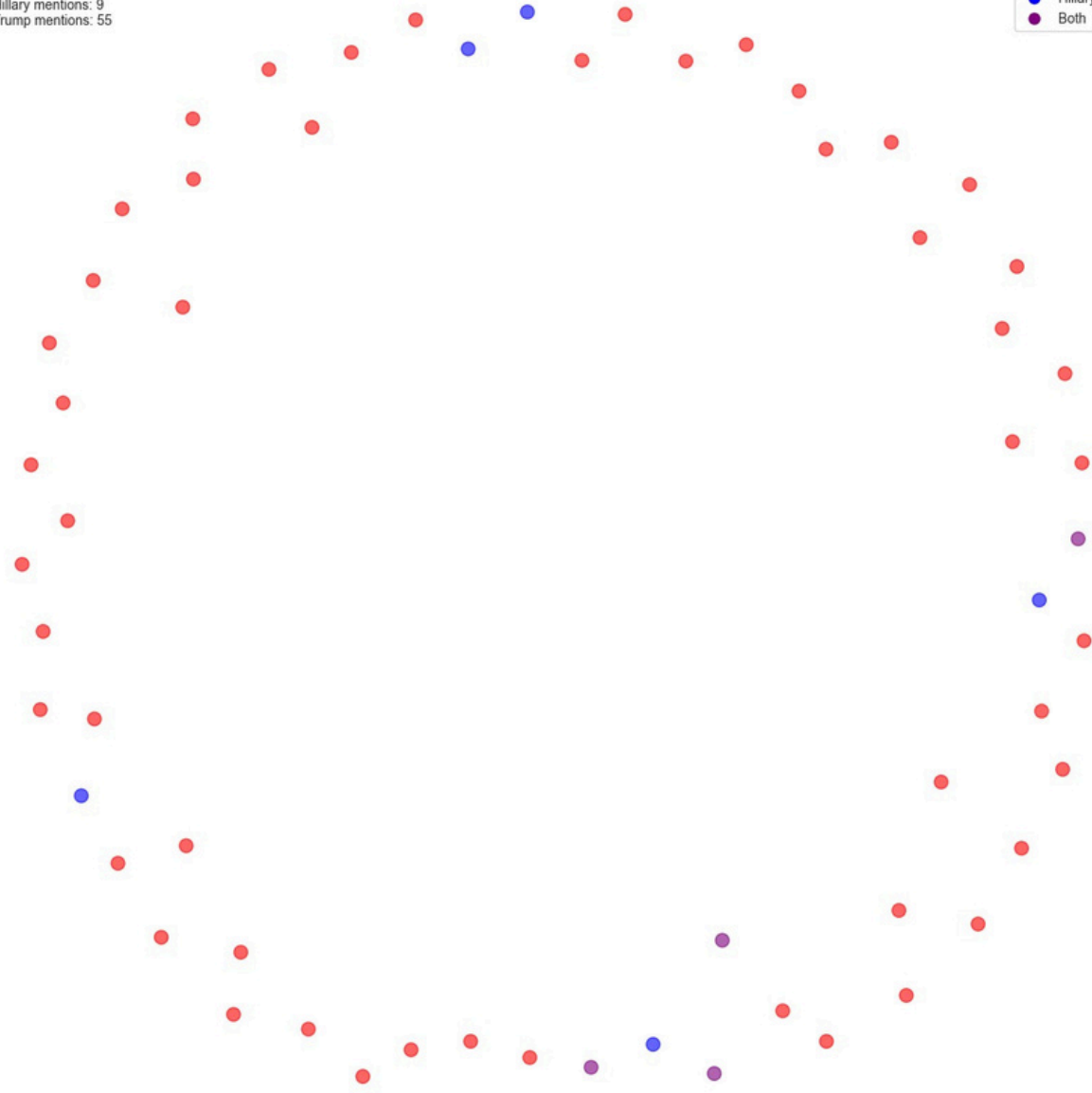
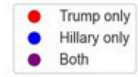


Mentions of Presidential Candidates Over Time



User Mention Network - RNC Convention

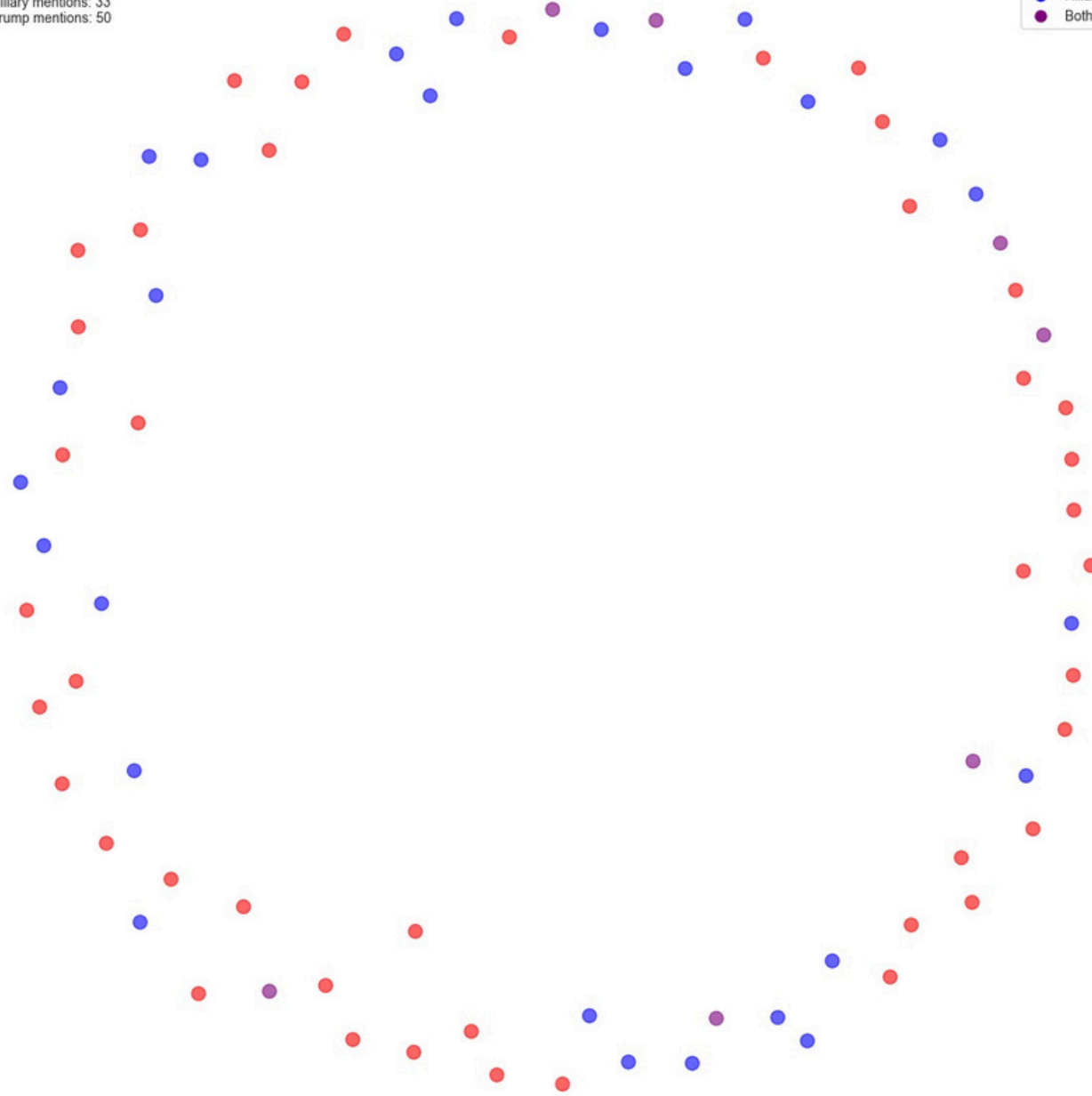
Total users: 60
Hillary mentions: 9
Trump mentions: 55



User Mention Network - DNC Convention

Total users: 76
Hillary mentions: 33
Trump mentions: 50

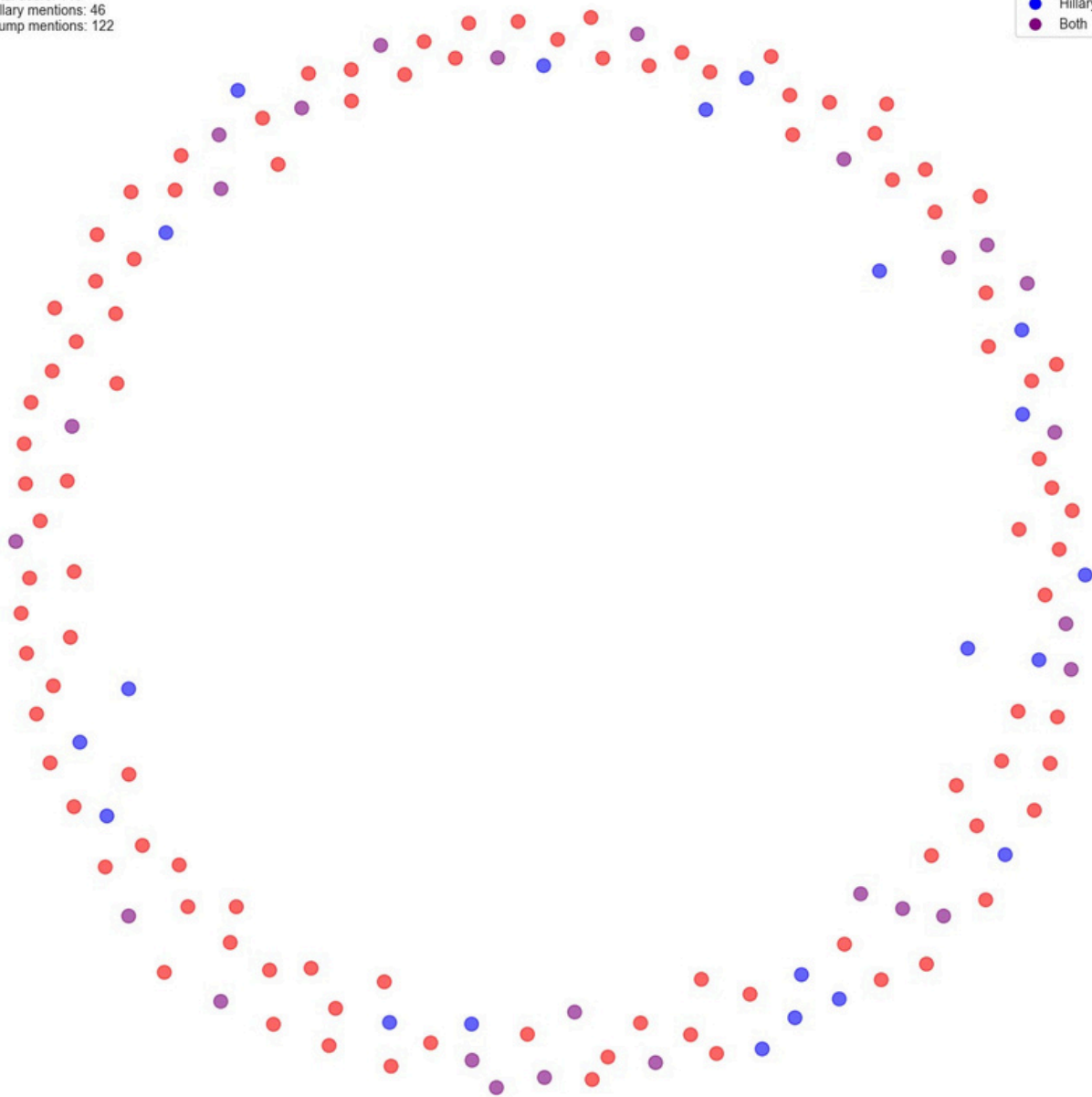
● Trump only
● Hillary only
● Both



User Mention Network - Election Day

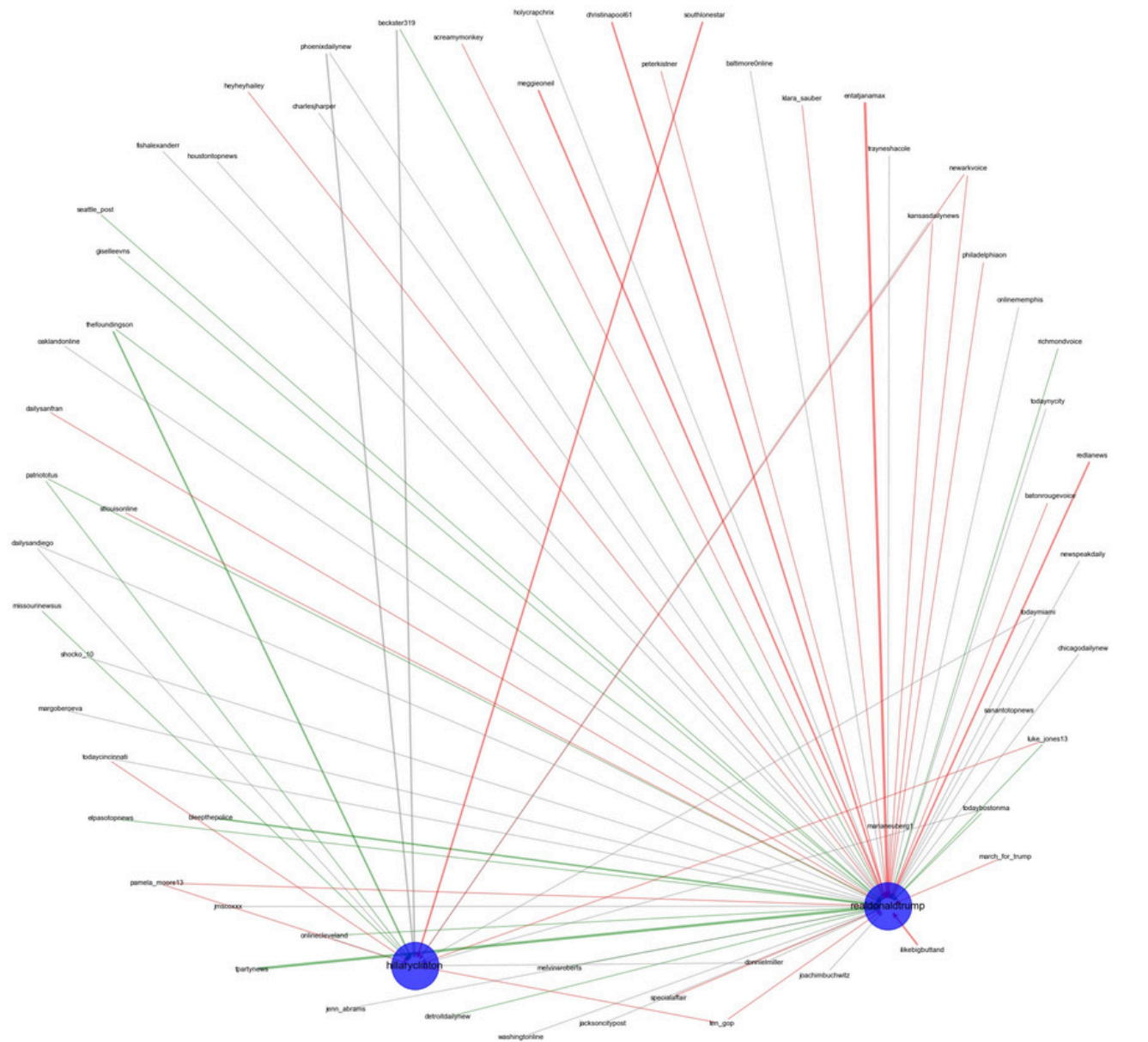
Total users: 143
Hillary mentions: 46
Trump mentions: 122

- Trump only
- Hillary only
- Both



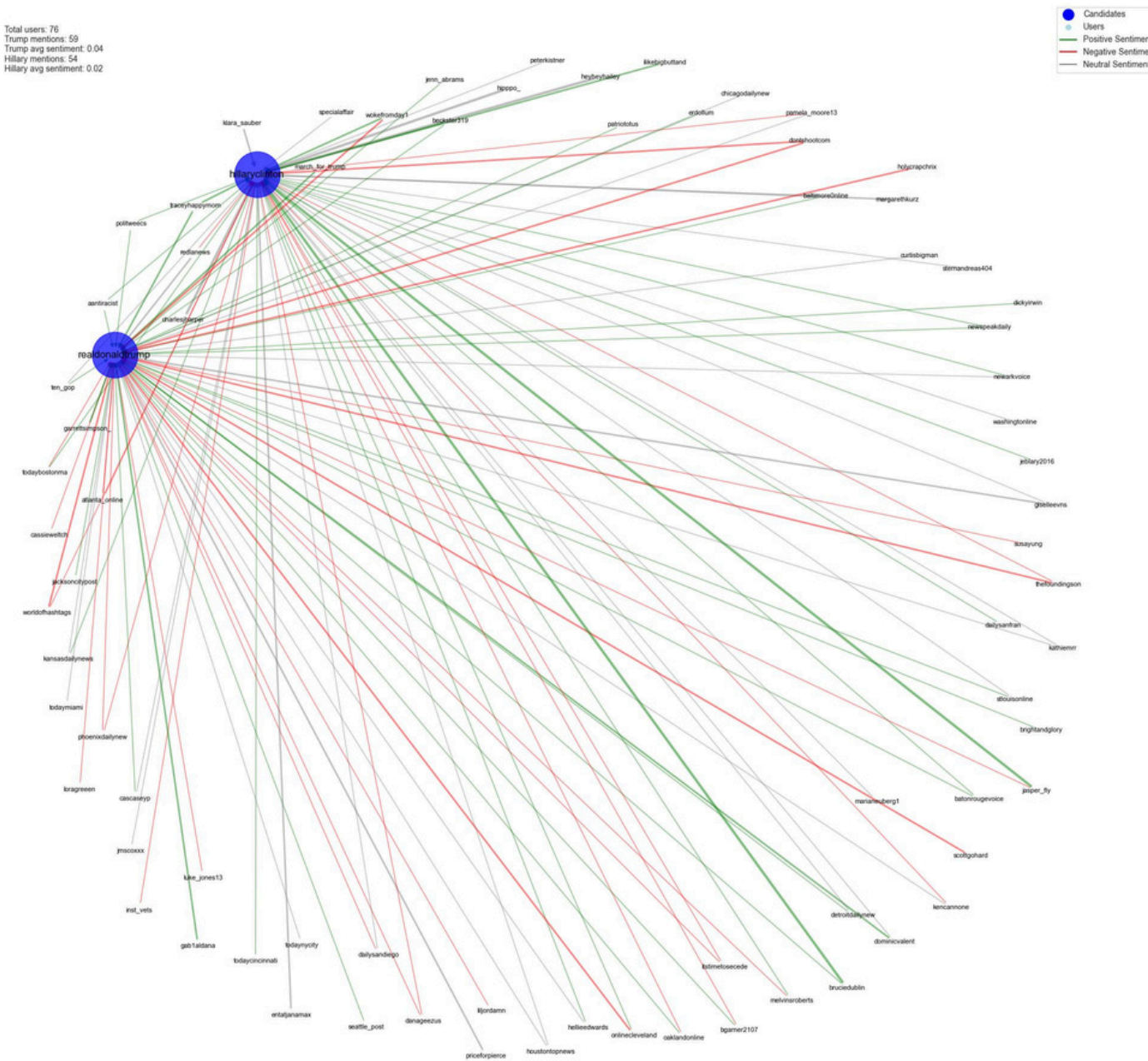
Mention Network with Sentiment - RNC Convention
2016-07-21

Total users: 60
Trump mentions: 58
Trump avg sentiment: -0.04
Hillary mentions: 17
Hillary avg sentiment: -0.13



Detailed Statistics for RNC Convention (2016-07-21):
Number of users: 60
Average sentiment towards Trump: -0.045
Average sentiment towards Hillary: -0.133

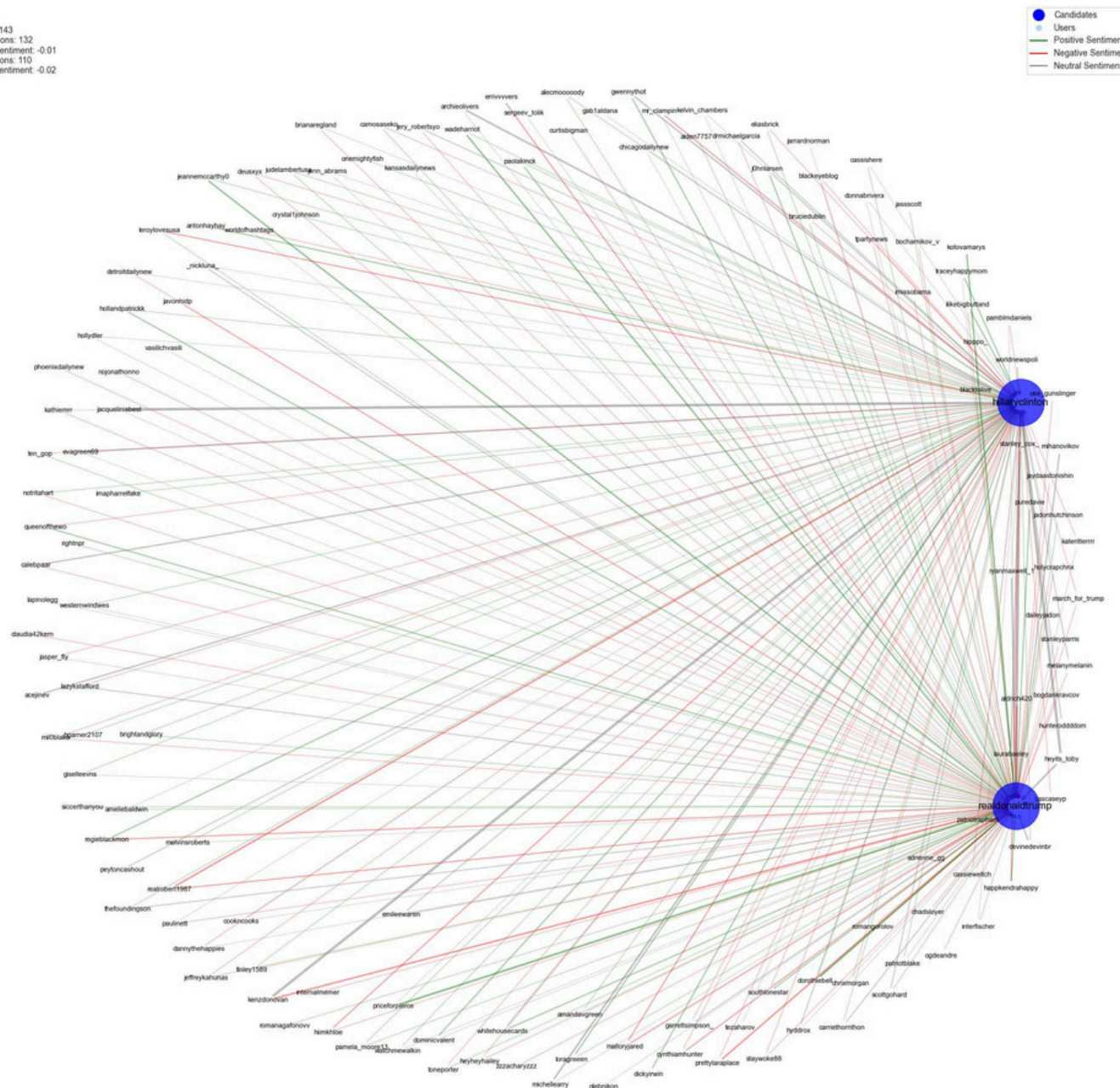
Total users: 76
Trump mentions: 59
Trump avg sentiment: 0.04
Hillary mentions: 54
Hillary avg sentiment: 0.02



```
Detailed Statistics for DNC Convention (2016-07-28):
Number of users: 76
Average sentiment towards Trump: 0.035
Average sentiment towards Hillary: 0.020
```

Mention Network with Sentiment - Election Day 2016-11-08

Total users: 143
Trump mentions: 132
Trump avg sentiment: -0.01
Hillary mentions: 110
Hillary avg sentiment: -0.02



Detailed Statistics for Election Day (2016-11-08):
Number of users: 143
Average sentiment towards Trump: -0.013
Average sentiment towards Hillary: -0.015

Russian accounts

Detailed Statistics:

Number of Russian users: 6

Total mentions of Trump: 29

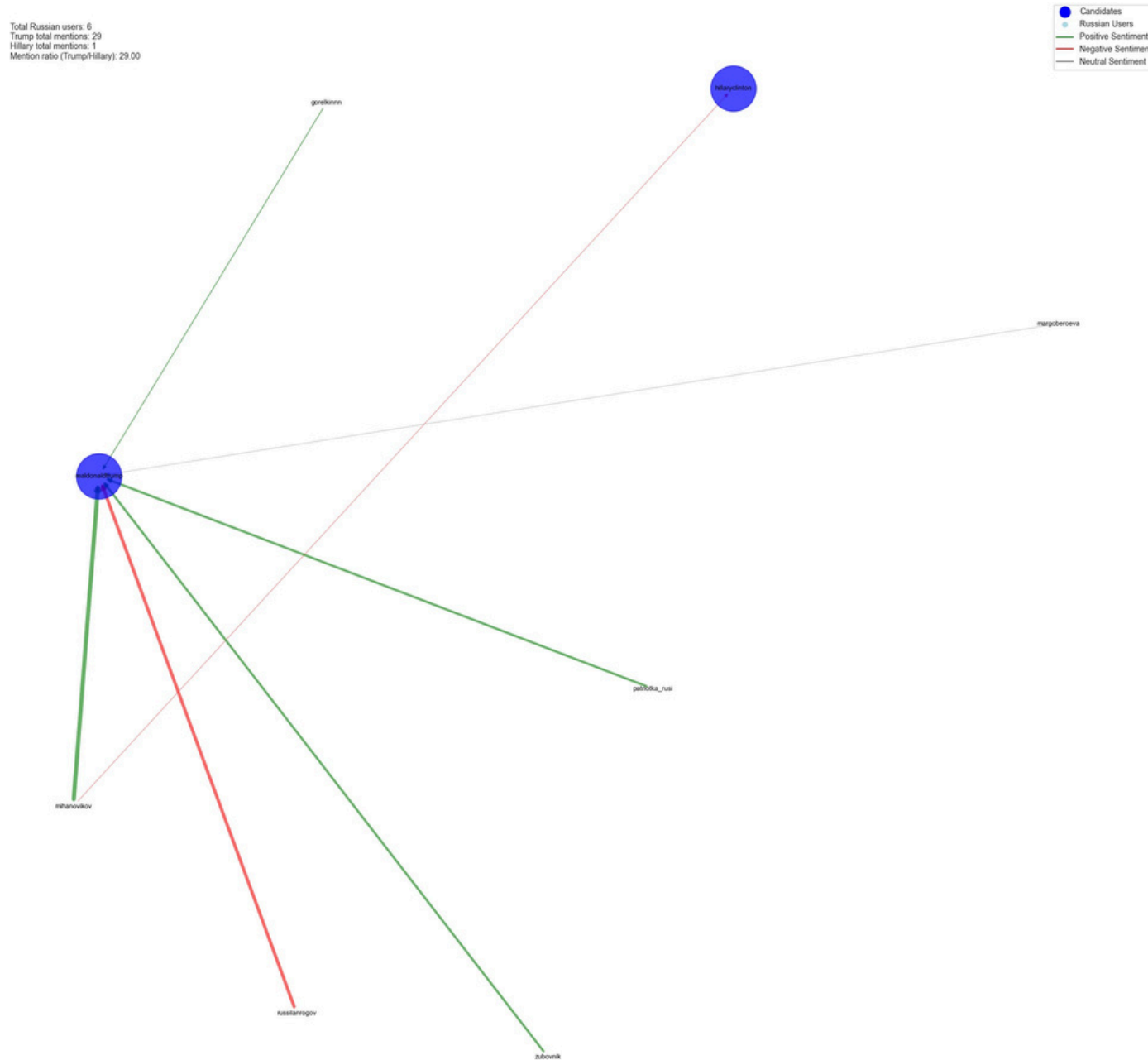
Total mentions of Hillary: 1

Average sentiment towards Trump: 0.087

Average sentiment towards Hillary: -0.477

Total Russian users: 6
Trump total mentions: 29
Hillary total mentions: 1
Mention ratio (Trump/Hillary): 29.00

Russian Users' Mentions of Presidential Candidates
Entire Timeline



The background features abstract, overlapping green geometric shapes, primarily triangles and polygons, in various shades of green, creating a modern, layered effect on the right side of the slide.

How to distinguish trolls?

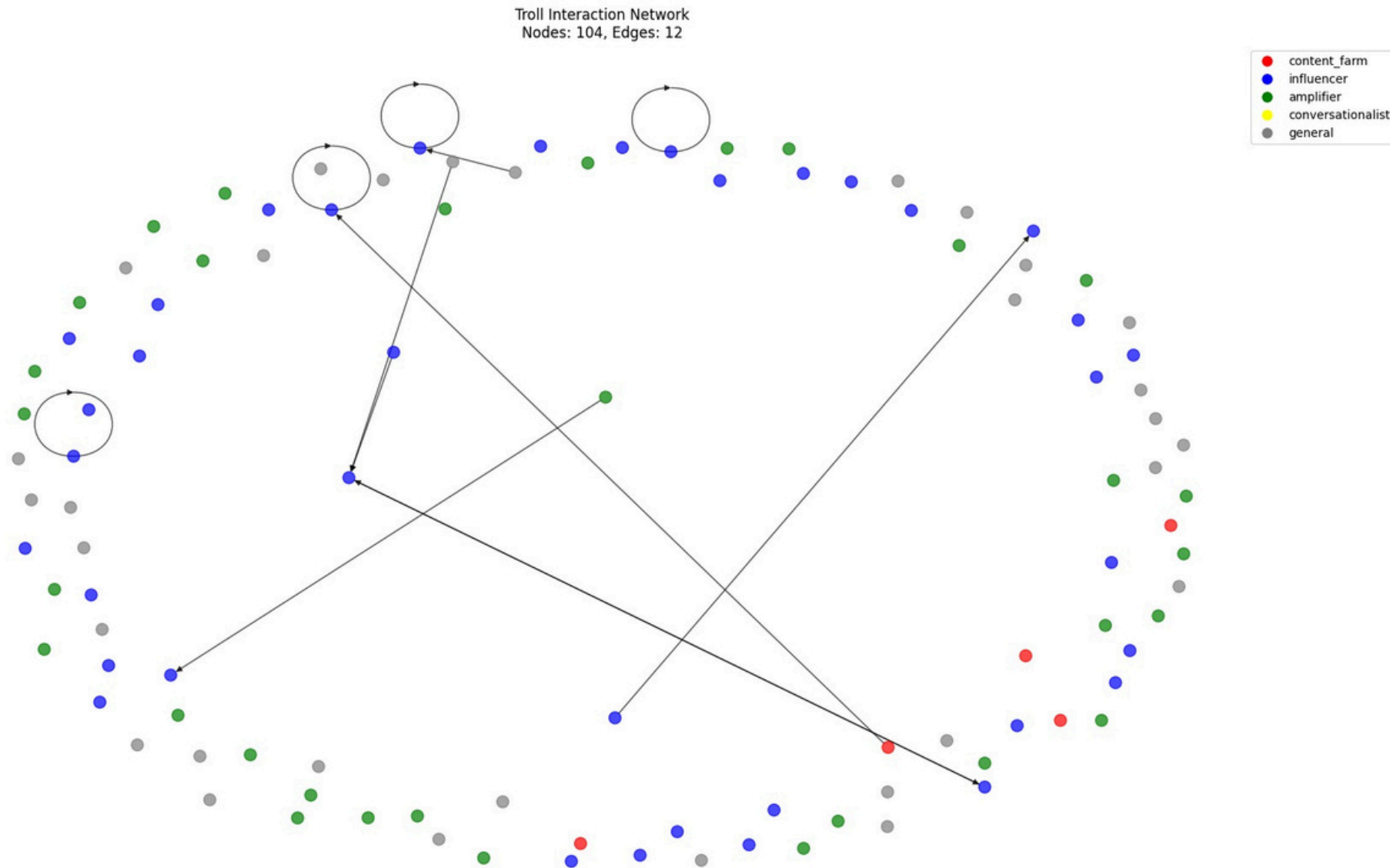
```
# Categorize based on behavior
if user['statuses_count'] > 1000 and user['followers_count'] < 100:
    troll_types['content_farms'].append(user['screen_name'])

if user['followers_count'] > 1000 and user['listed_count'] > 10:
    troll_types['influencers'].append(user['screen_name'])

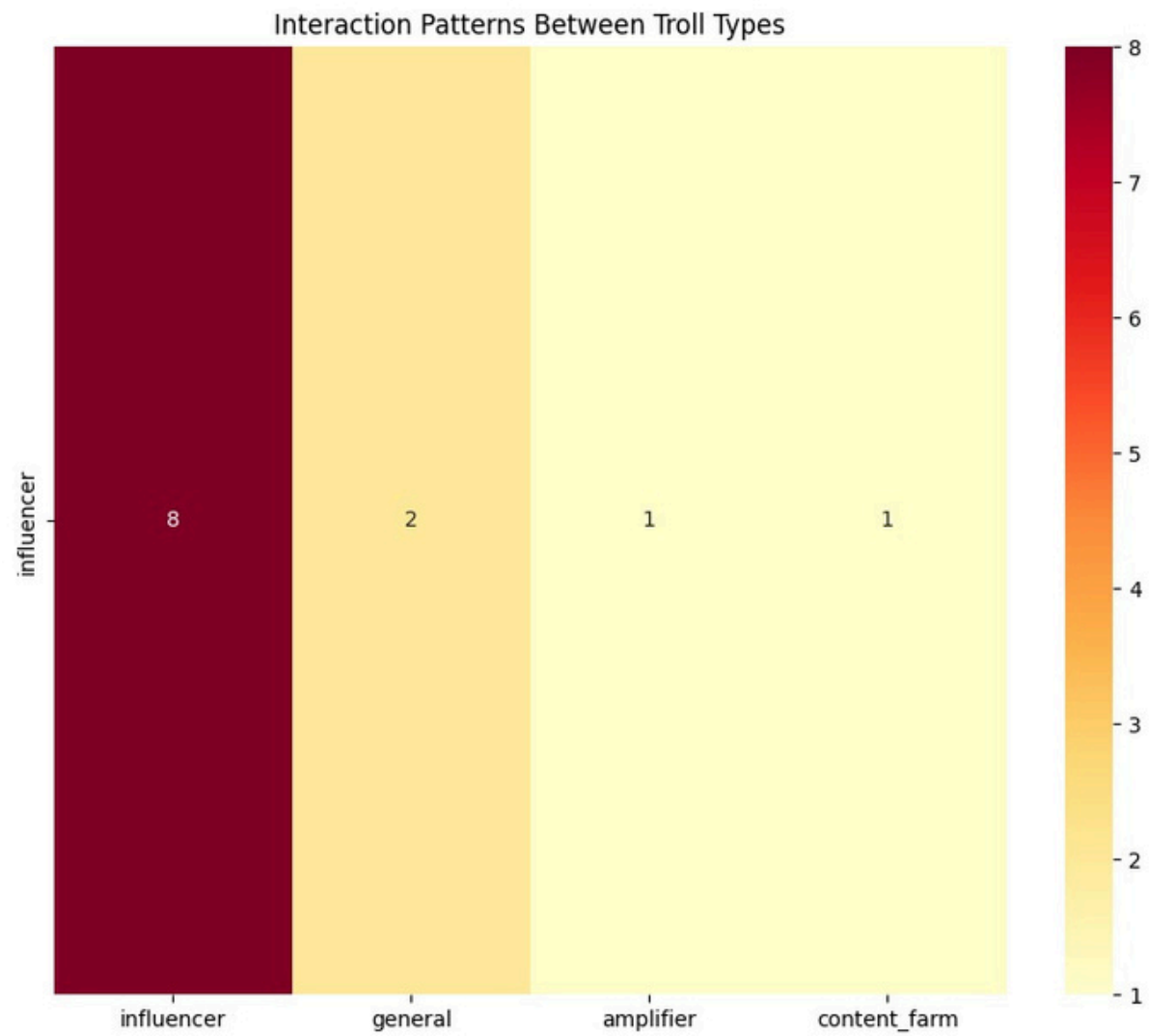
if retweet_ratio > 0.7:
    troll_types['amplifiers'].append(user['screen_name'])

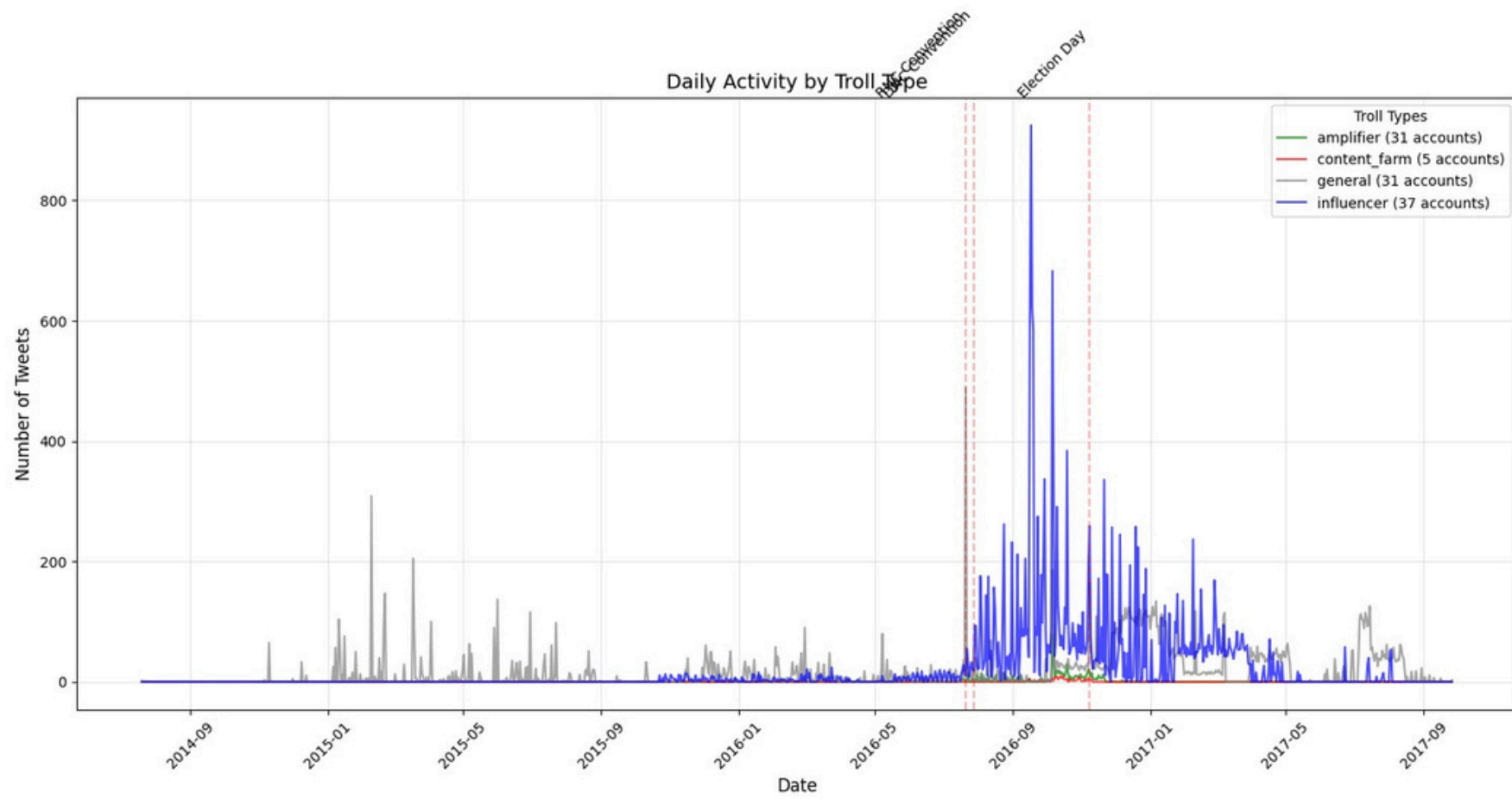
if reply_ratio > 0.5:
    troll_types['conversationalists'].append(user['screen_name'])
```

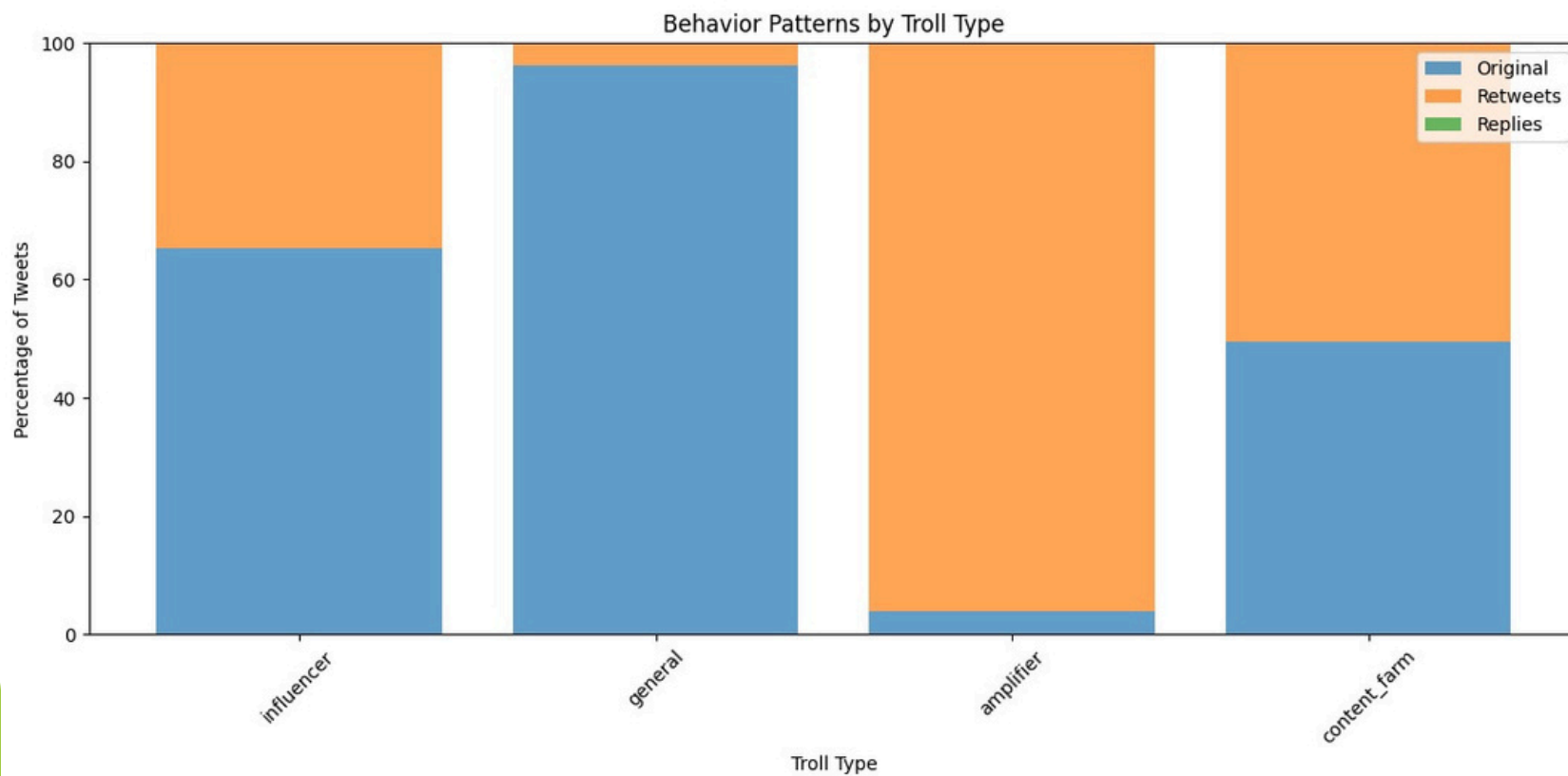
General == conversationalist



Troll type distribution:
general: 31
content_farm: 5
influencer: 37
amplifier: 31





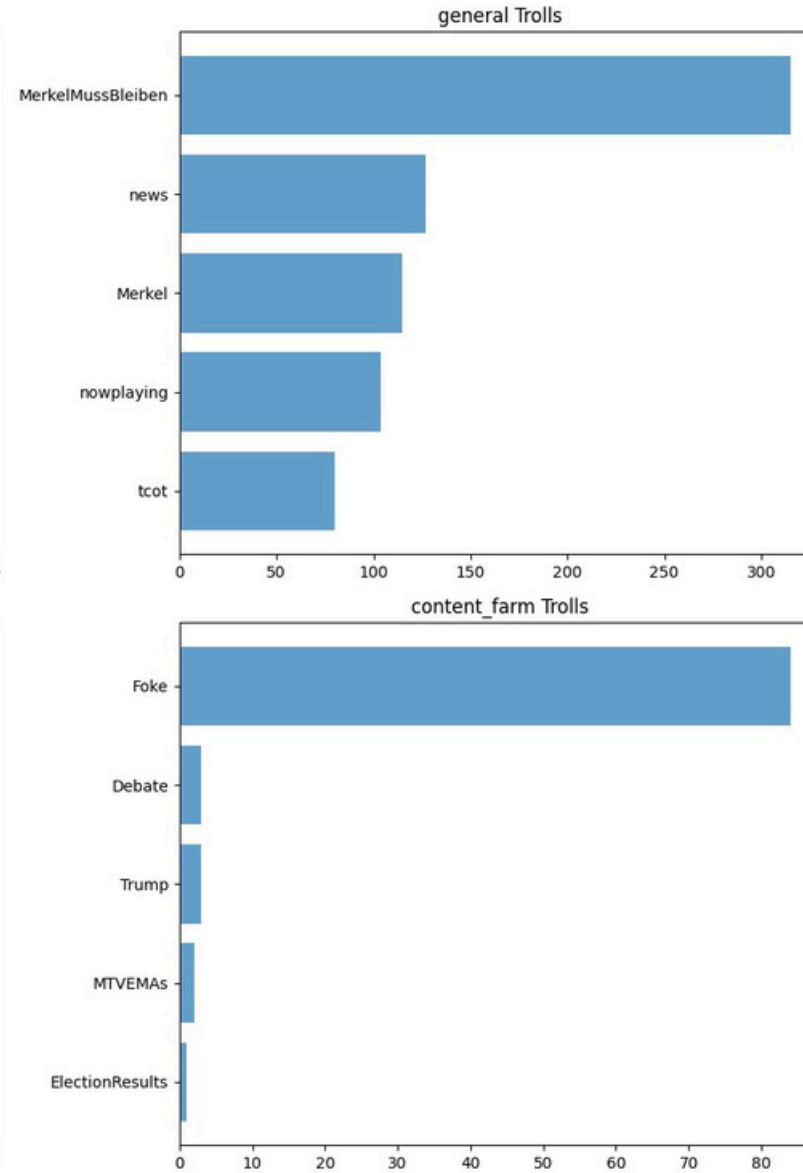
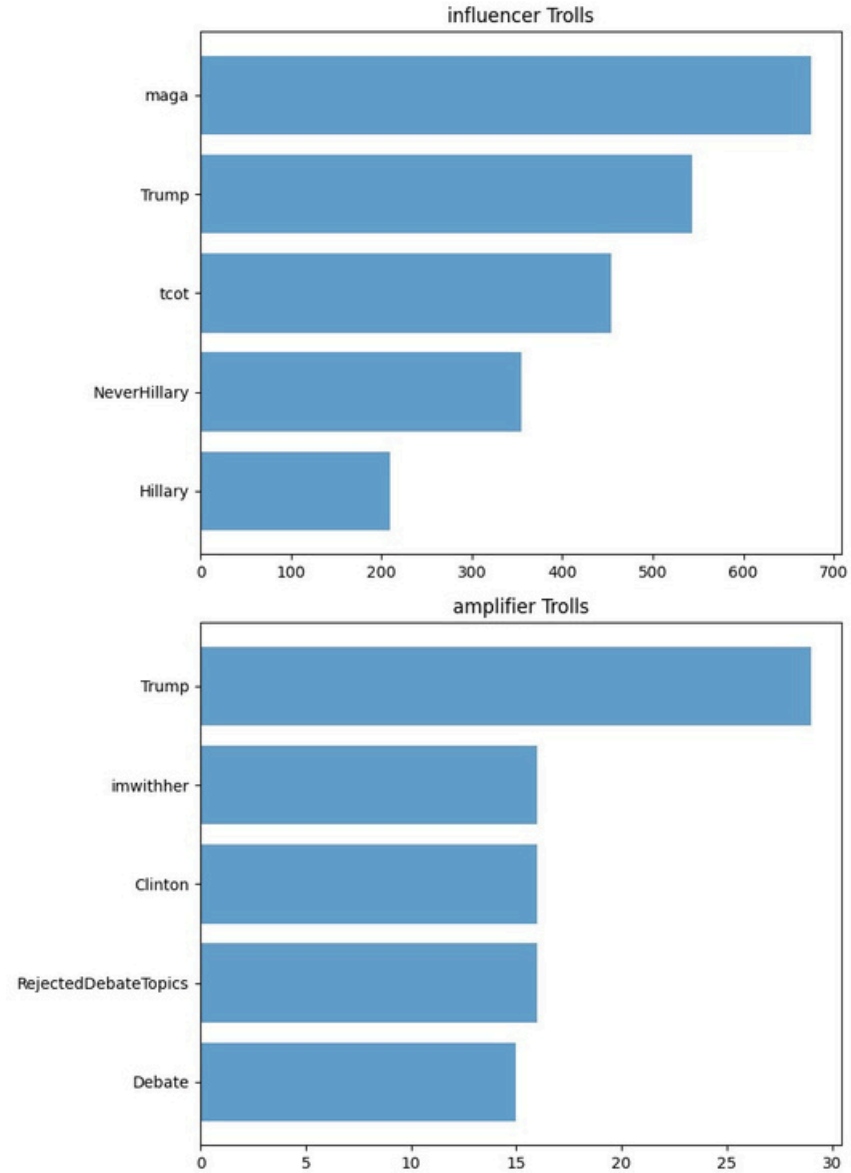


INFLUENCER TROLLS:
Total tweets: 22715
Tweet composition:
- Original: 65.2%
- Retweets: 34.7%
- Replies: 0.1%

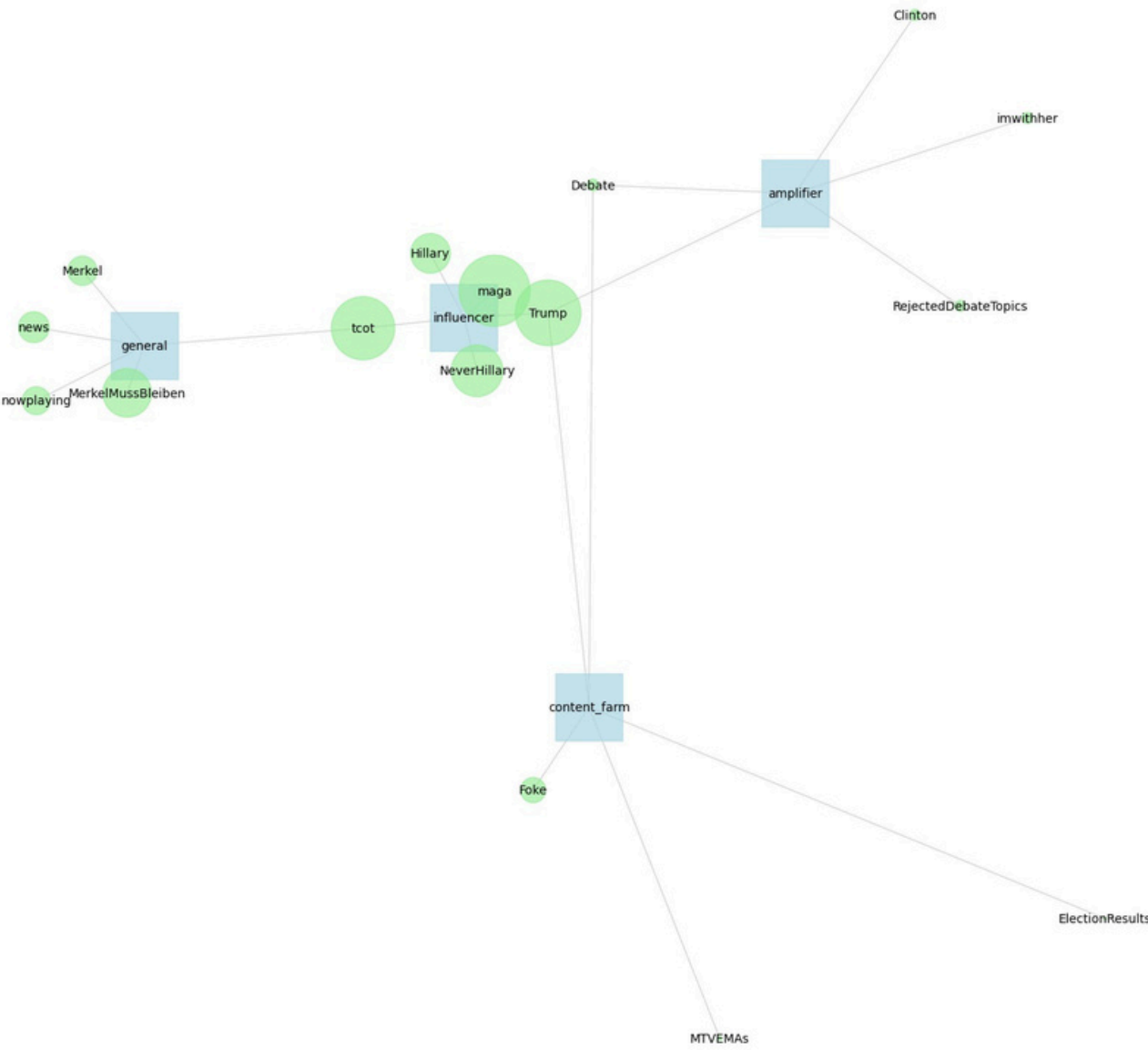
Top hashtags:
- maga: 675
- Trump: 543
- tcot: 454
- NeverHillary: 355
- Hillary: 210

GENERAL TROLLS:
Total tweets: 19115
Tweet composition:
- Original: 96.2%
- Retweets: 3.8%
- Replies: 0.0%

Top 5 Hashtags by Troll Type



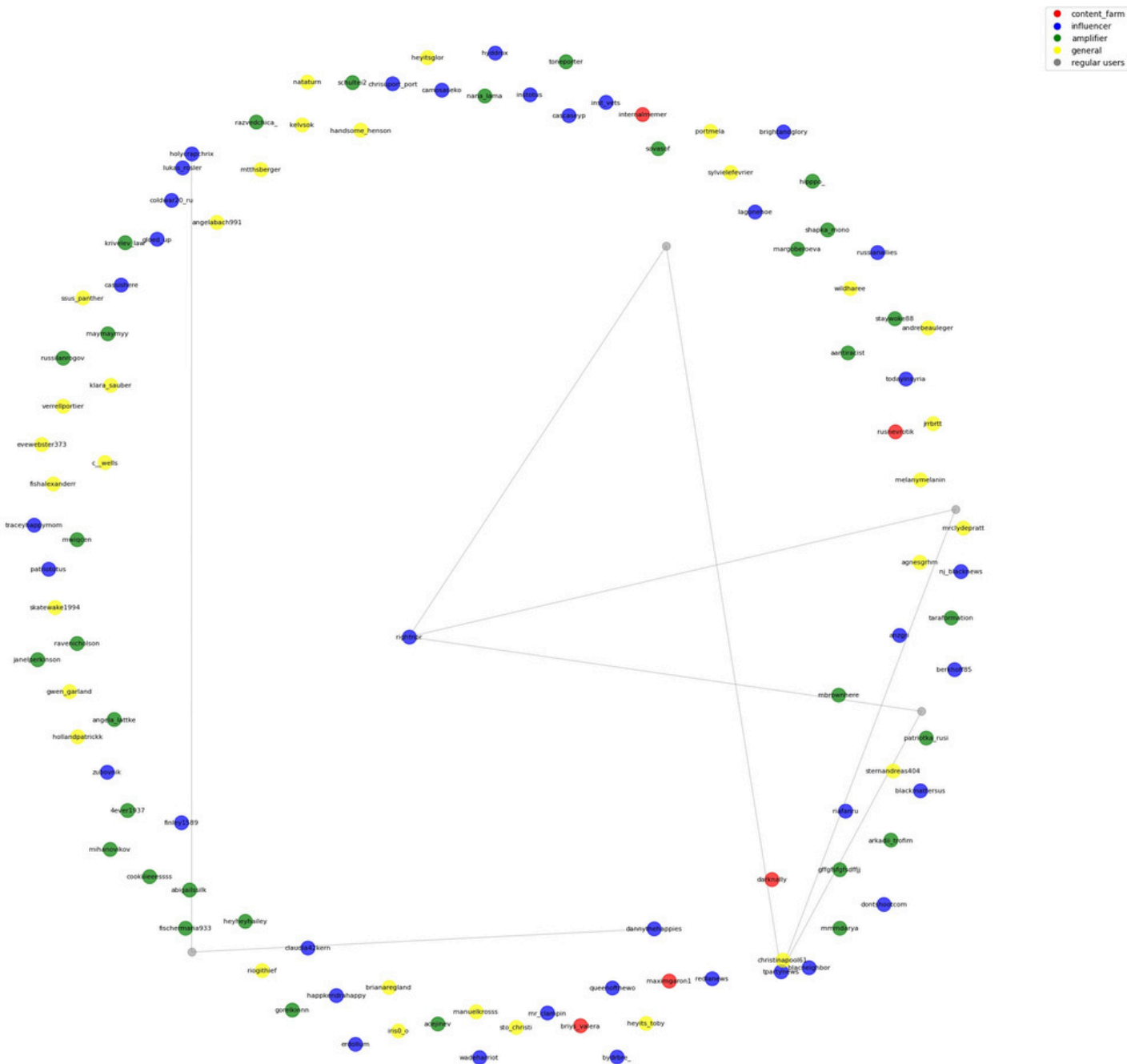
Hashtag Usage by Troll Type
(Node size represents frequency of usage)



The background features abstract, overlapping green geometric shapes, primarily triangles and polygons, in various shades of green, creating a modern, layered effect on the right side of the slide.

Trolls masking in the society

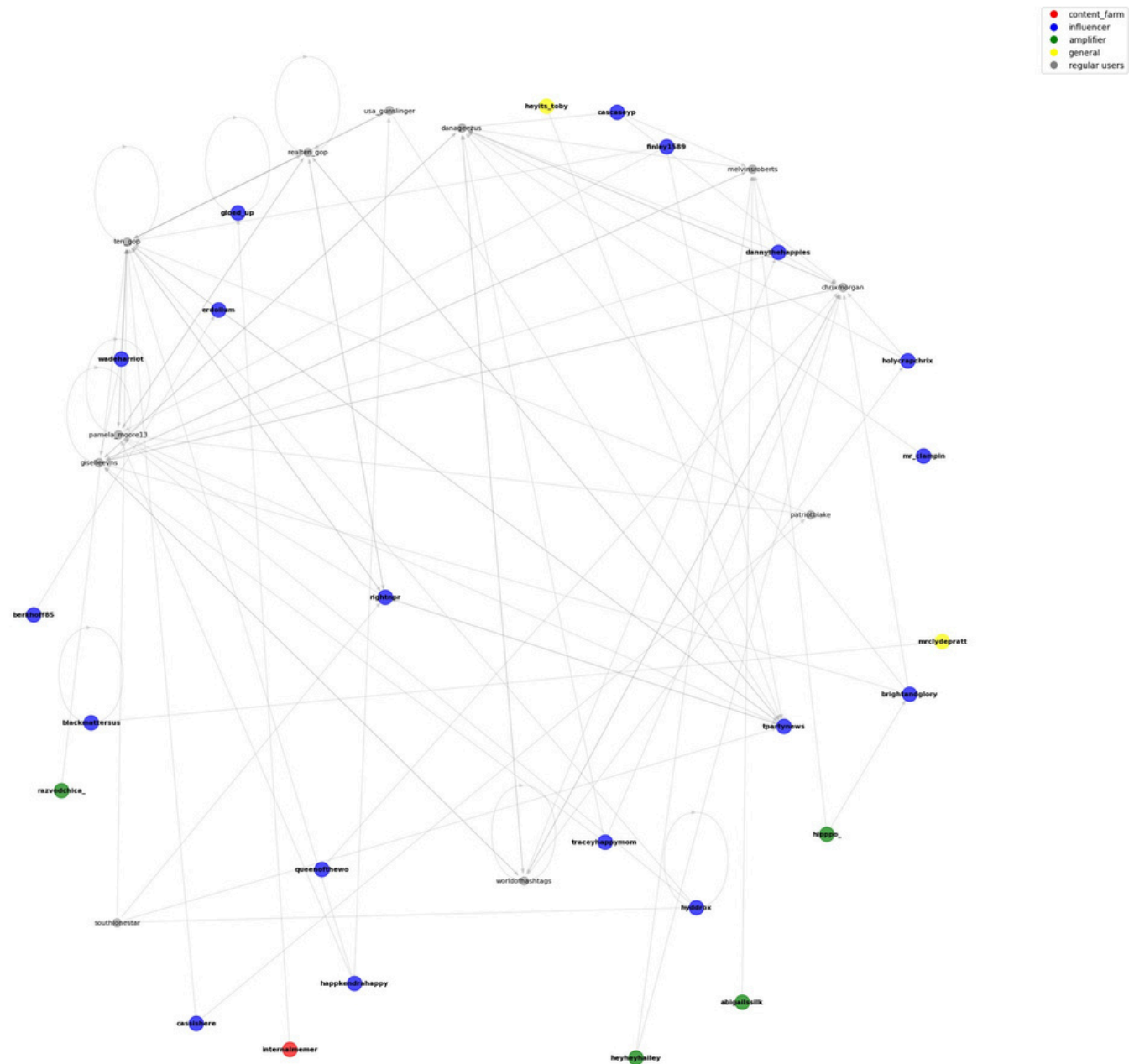
Troll Network Through Common Mentions
(Trolls connected through users who mention them)



Network Statistics:
Number of nodes: 108
Number of edges: 8

Number of nodes by type:
general: 31
content_farm: 5
influencer: 37
amplifier: 31
regular_users: 4

Troll Communities Through Mention Chains
(Directed paths showing mention flow)



Total paths found: 142
Average path length: 3.72